

NEET(UG)-2024 (MADHOPUR EXAMINATION)

(Held On Sunday 5th MAY, 2024)

PHYSICS

TEST PAPER WITH ANSWER AND SOLUTION

Physics : Section-A (Q. No. 1 to 35)

1. Match List-I with List-II

List-I (Application of Gauss Law)	List-II (Value of $ E $)
A. Field inside thin shell	I. $\frac{\lambda}{2\pi\epsilon_0 r} \hat{n}$
B. Field outside thin shell	II. $\frac{q}{4\pi\epsilon_0 R^2} \hat{r}$
C. Field of thin shell at the surface	III. $\frac{q}{4\pi\epsilon_0 r^2} \hat{r}$
D. Field due to long charged wire	IV. Zero

(Here symbols have their usual meaning and R is the radius of thin shell)

Choose the correct answer from the options given below :

- (1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-IV, B-III, C-I, D-II

Ans. (2)

Sol. A : $E = 0$

$$B : E = \frac{kq}{r^2} \hat{r}$$

$$C : E = \frac{kq}{R^2} \hat{r}$$

$$D : E = \frac{2k\lambda}{r} \hat{r}$$

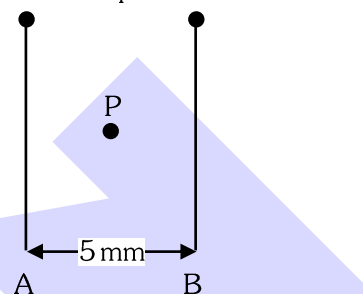
2. A particle is displaced through $(3\hat{i} + 4\hat{j})$ m by a force $2\hat{i}$ N. The work done is :

- (1) 8 J
(2) 6 J
(3) 10 J
(4) 14 J

Ans. (2)

$$\begin{aligned} \text{Sol. } W &= \vec{F} \cdot \vec{S} \\ &= 2(3) + 0 \\ &= 6 \text{ J} \end{aligned}$$

3. Two identical rectangular plane sheets A and B each of surface charge density $\epsilon_0 \text{Cm}^{-2}$ are placed parallel to each other as shown in figure. The electric field at the mid point P will be :



- (1) 1 N C^{-1} (2) 0.5 N C^{-1}
(3) Zero (4) 2 N C^{-1}

Ans. (3)

$$\text{Sol. } \vec{E} = \vec{E}_{P/A} + \vec{E}_{P/B}$$

$$\vec{E} = E(\hat{i}) + E(-\hat{i})$$

$$E = 0$$

4. An object falls freely from height h above the ground. It travels $\frac{5}{9}h$ in the last 1s. Then height h is

(Given $g = 10 \text{ ms}^{-2}$)

- (1) 25 m (2) 45 m (3) 58 m (4) 5 m

Ans. (2)

$$\text{Sol. } t = \sqrt{\frac{2h}{g}} \text{ and } t-1 = \sqrt{\frac{2(4h)}{g}}$$

$$\Rightarrow h = 45 \text{ m}$$

5. Radiation of wavelength 280 nm is used in an experiment of photoelectric effect with cathode of work function, 25 eV. The maximum kinetic energy of the photoelectrons is

[Take $h = 6.62 \times 10^{-34} \text{ J s}$ and $c = 3 \times 10^8 \text{ ms}^{-1}$]

- (1) $7.103 \times 10^{-15} \text{ J}$ (2) 1.9 eV
(3) 4.60 eV (4) 4.4 eV

Ans. (2)

$$\text{Sol. } K_{\max} = \left[\frac{12400}{2800} - 2.5 \right] \text{ eV}$$

$$= (4.4 - 2.5) \text{ eV}$$

$$= 1.9 \text{ eV}$$



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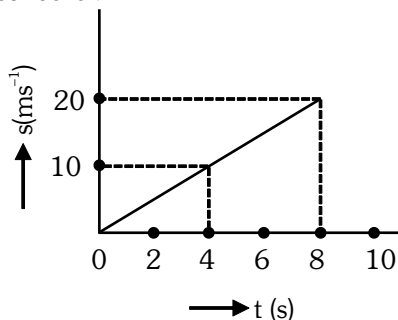
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6. The speed (s) of a car as a function of time (t) is shown in figure. The distance travelled by the car in 8 seconds is :



- (1) 60 m (2) 80 m
(3) 18 m (4) 180 m

Ans. (2)

Sol. Distance is area under speed time graph.

$$D = \frac{1}{2}(8)(20) = 80\text{m}$$

7. A conducting circular loop of face area $2.5 \times 10^{-3} \text{ m}^2$ is placed perpendicular to a magnetic field which varies as $B = 0.5 \sin(100\pi t)\text{T}$. The magnitude of induced emf at time $t = 0 \text{ s}$ is :

- (1) $125\pi \text{ mV}$ (2) $125\pi \text{ V}$
(3) $12.5\pi \text{ mV}$ (4) $0.125\pi \text{ mV}$

Ans. (1)

Sol. $\text{emf} = \left| -A \frac{dB}{dt} \right|$

$$= 2.5 \times 10^{-3} \times 0.5 \times 100\pi \cos(100\pi t)$$

$$= 125\pi \times 10^{-3}$$

$$= 125\pi \text{ mV}$$

8. A vernier calipers has 1 mm marks on the main scale. It has 20 equal divisions on the vernier scale which match with 16 main scale divisions. The least count of this vernier calipers is :

- (1) 0.05 mm
(2) 0.10 mm
(3) 0.20 mm
(4) 0.02 mm

Ans. (3)

Sol. $\text{LC} = \text{MSD} - \text{VSD}$

$$= \text{MSD} - \frac{16}{20} \text{ MSD}$$

$$= \frac{\text{MSD}}{5} = \frac{1}{5} \text{ mm}$$

9. Zener diode works
- (1) in reverse bias only
 - (2) as a voltage regulator in forward bias and as a simple pn junction diode in reverse bias.
 - (3) as a voltage regulator in reverse bias and as a simple pn junction diode in forward bias.
 - (4) in forward bias only.

Ans. (3)

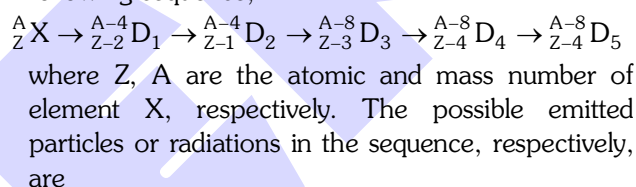
10. For an adiabatic process, the factor which remains constant is [All the notations have their usual meaning] :

- (1) $TV^{1-\gamma}$ (2) $V^{\gamma-1}T^{\gamma}$
(3) $P^{1-\gamma}T^{\gamma}$ (4) $P^{\gamma-1}T^{\gamma}$

Ans. (3)

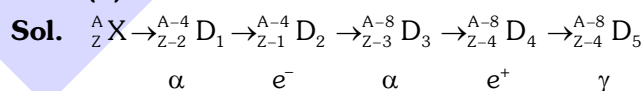
Sol. Equation of state : $P^{1-\gamma}T^{\gamma} = \text{constant}$

11. An excited heavy nucleus ${}^A_Z\text{X}$ emits radiations in the following sequence,



- (1) e^- , α , e^+ , α , γ
(2) α , e^- , α , e^+ , γ
(3) α , e^+ , α , e^- , γ
(4) e^+ , α , e^- , α , γ

Ans. (2)



12. Time taken by sunlight to pass through a glass slab of thickness 5 mm and refractive index 1.5, is :

- (1) $\left(\frac{5}{2}\right) \times 10^{-11} \text{ s}$
(2) $\left(\frac{5}{3}\right) \times 10^{-11} \text{ s}$
(3) $\left(\frac{5}{2}\right) \times 10^{-8} \text{ s}$
(4) $\left(\frac{5}{3}\right) \times 10^{-8} \text{ s}$

Ans. (1)

Sol. $t = \frac{d}{C/\mu} = \frac{5 \times 10^{-3}}{3 \times 10^8} \times 1.5$

$$= \frac{5}{2} \times 10^{-11}$$



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13. Given below two statements :

Statement-I : The de Broglie wavelength associated with a material particle depends on its charge and nature.

Statement-II : The wave nature of particles, in sub-atomic domain is significant and measurable.

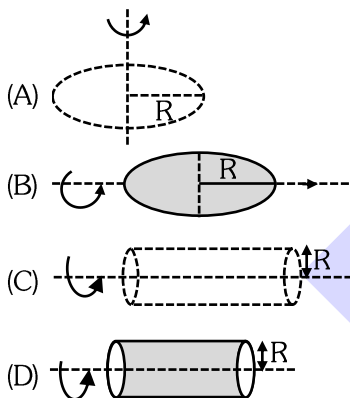
In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) **Statement I** is correct but **Statement II** is incorrect.
- (3) **Statement I** is incorrect but **Statement II** is correct.
- (4) Both **Statement I** and **Statement II** are correct.

Ans. (3)

Sol. Statement-1 is wrong because de-Broglie wavelength is associated with all types of material particle whether they are charged or uncharged. Statement-2 correct option (3)

14. Consider a thin circular ring (A), a circular disc (B), a hollow cylinder (C) and a solid cylinder (D) of the same radius R and of same mass.



If I_A , I_B , I_C and I_D are their moments of inertia about the axis shown, then :

Choose the **correct** answer from the options given below :

- (1) $I_A = 2I_B$ and $2I_C = I_D$
- (2) $2I_A = I_C$ and $I_B = 2I_D$
- (3) $I_A = I_B = I_C = 2I_D$
- (4) $I_A = I_C$ and $2I_B = I_D$

Ans. (4)

Sol. $I_A = mR^2$
 $I_B = mR^2/4$
 $I_C = mR^2$
 $I_D = mR^2/2$
 $\Rightarrow I_A = I_C$ & $2I_B = I_D$

15. In Young's double slit experiment, if the wavelength of light used is increased (say from violet to red) then the :

- (1) fringe width increases
- (2) central bright fringe becomes dark
- (3) fringe width remains unaltered.
- (4) fringe width decreases.

Ans. (1)

Sol. In YDSE

$$\text{Fringe width} : \beta = \frac{\lambda D}{d}$$

If $\lambda \uparrow$ then β also $\uparrow$

16. Given below two statements: One is labelled as

Assertion A and the other is labelled as **Reason R**.

Assertion A: A glass tube partially filled with water represents an open organ pipe.

Reason R : The open end corresponds to an antinode and the end in contact with water, to a node.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (2) **A** is true but **R** is false.
- (3) **A** is false but **R** is true.
- (4) Both **A** and **R** are true but **R** is the correct explanation of **A**.

Ans. (3)

Sol. Assertion is false as water acts as fixed end (for displacement)

Reason is true for displacement.

17. Match **List-I** with **List-II**

List I	List II
(Material)	(Example)
A. Diamagnet	I. Alnico
B. Paramagnet	II. Copper
C. Soft ferromagnet	III. Aluminium
D. Hard ferromagnet	IV. Gadolinium

Choose the **correct** answer from the options given below :

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-I, D-IV
- (3) A-II, B-III, C-IV, D-I
- (4) A-I, B-II, C-III, D-IV

Ans. (3)



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Sol. Diamagnet → Copper

Paramagnet → Aluminium

Soft ferromagnet → Gadolinium

Hard ferromagnet → Alnico

18. If $\vec{E}$ and $\vec{B}$ represents the electric field vector and magnetic field vector, respectively, in an electromagnetic wave then the direction of EM wave is along:

- (1) $\vec{B}$ (2) $\vec{E} \times \vec{B}$
(3) $\vec{B} \times \vec{E}$ (4) $\vec{E}$

Ans. (2)

Sol. $\hat{S} = \hat{E} \times \hat{B}$

19. A wire of length L and radius r ($r \ll L$) is kept floating on the surface of a liquid of density ρ . The maximum radius of the wire for which it may not sink is (Surface tension of liquid is T) :

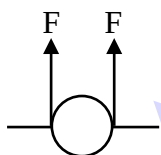
- (1) $\sqrt{\frac{2T}{\rho g}}$ (2) $\sqrt{\frac{2T\rho}{\pi g}}$
(3) $\sqrt{\frac{2T}{\pi\rho g}}$ (4) $\sqrt{\frac{T}{\rho g}}$

Ans. (3)

Sol. $2F = mg$

$$2TL = \rho\pi r^2 Lg$$

$$r = \sqrt{\frac{2T}{\rho\pi g}}$$



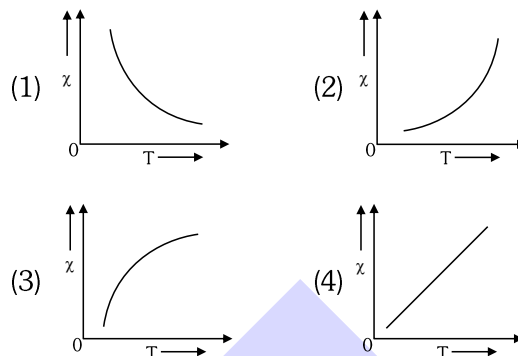
20. L , C and R represent the value of inductance, capacitance and resistance, respectively. The factor which has the same dimensions as that of the inverse of resonance frequency is :

- (1) $\sqrt{\frac{L}{C}}$ (2) $\frac{C}{L}$
(3) $\frac{R}{L}$ (4) $\sqrt{LC}$

Ans. (4)

Sol. $\omega = \frac{1}{\sqrt{LC}}$

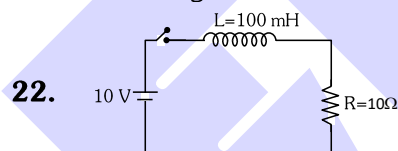
21. The best suited curve showing the variation of susceptibility (χ) of a paramagnetic material in free space with temperature (T) is :



Ans. (1)

Sol. $\chi \propto \frac{1}{T}$

According to Curie Law



22.

An ideal inductor-resistor-battery circuit is switched on at $t=0$ s. At time t , the current is $i=i_0\left(1-e^{-\frac{t}{\tau}}\right)$ A,

where i_0 is the steady-state value. The time at which the current becomes $0.5 i_0$ is : [Given $\ln(2)=0.693$]

- (1) 6.93 ms (2) 69.3 s
(3) 6.93 s (4) 6.93×10^3 s

Ans. (1)

Sol. $i = 0.5 i_0$

$$\Rightarrow t_{1/2} = 0.693 \frac{L}{R}$$

$$\Rightarrow t_{1/2} = 0.693 \times \frac{100 \times 10^{-3}}{10} = 6.93 \text{ ms}$$

23. If $F = \alpha t^2 - \beta t$ is the magnitude of the force acting on a particle at an instant t then the time, at which the force becomes constant, is (where α and β are constants) :

- (1) $\frac{\beta}{2\alpha}$ (2) $\frac{2\beta}{\alpha}$ (3) zero (4) $\frac{\beta}{\alpha}$

Ans. (1)



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Sol. Force is constant

$$\text{when } \frac{dF}{dt} = 0$$

$$\Rightarrow \frac{d}{dt}(\alpha t^2 - \beta t) = 0$$

$$\Rightarrow 2\alpha t - \beta = 0$$

$$\Rightarrow \tau = \beta/2\alpha$$

24. The percentage errors in the measurement of mass and momentum of an object are 1% and 2% respectively. The percentage error in the measurement of kinetic energy of the object is :

- (1) 3%
- (2) 4%
- (3) 5%
- (4) 1%

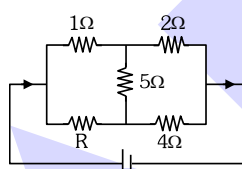
Ans. (3)

Sol. $KE = \frac{p^2}{2m}$

$$\Rightarrow \frac{\Delta KE}{KE} \% = \frac{\Delta m}{m} \% + \frac{2\Delta p}{p} \%$$

$$= 1\% + 2(2\%) = 5\%$$

25. The value of R in the given circuit when there is no current in the 5Ω resistor is :



- (1) 9Ω
- (2) 3Ω
- (3) 2Ω
- (4) 12Ω

Ans. (3)

Sol. Balanced wheatston bridge : $\frac{P}{Q} = \frac{R}{S}$

$$\frac{1}{2} = \frac{R}{4} \Rightarrow R = 2\Omega$$

26. Match **List-I** with **List-II** :

List - I (Measured values)	List - II (Significant figures)
---	--

A. 0.001213	I. 2
B. 2.1×10^{16}	II. 3
C. 3.70	III. 1
D. 3000	IV. 4

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II, D-IV (2) A-I, B-II, C-IV, D-III
(3) A-IV, B-I, C-II, D-III (4) A-III, B-II, C-I, D-IV

Ans. (3)

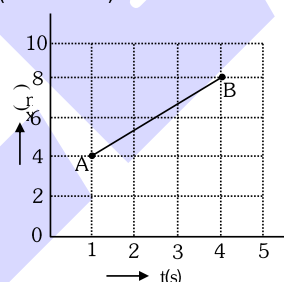
Sol. $0.001213 \rightarrow 4$

$$2.1 \times 10^{16} \rightarrow 2$$

$$3.70 \rightarrow 3$$

$$3000 \rightarrow 1$$

27. A body of mass 6 kg is moving from its initial position A to the next position B as shown in figure. From A to B, the value of momentum of the body is (in SI unit)



- (1) 12 (2) 8 (3) 6 (4) 24

Ans. (2)

Sol. $p = mv$

$$= m \left(\frac{dx}{dt} \right)$$

$$= 6 \frac{(8-4)}{4-1} = 8 \text{ SI units}$$

28. A solar cell is a :

- (1) type of light emitting diode.
- (2) diode working on the principle of photo - voltaic effect
- (3) type of photo-diode with external biasing.
- (4) forward biased zener diode.

Ans. (2)



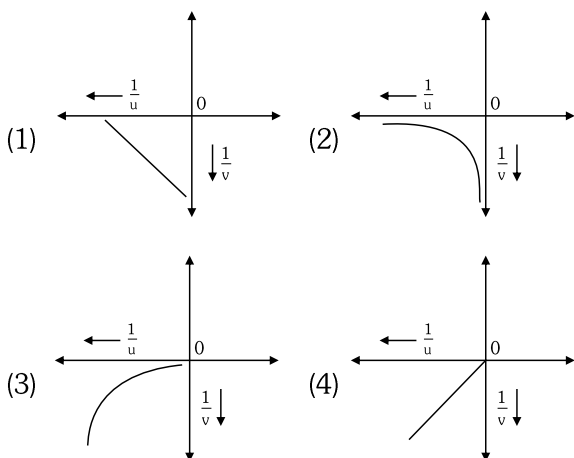
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29. The graph which shows the correct variation of $\frac{1}{v}$ with $\frac{1}{u}$ for a concave mirror, where u is the object distance and v is the image distance, is :

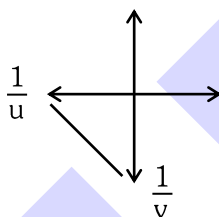


Ans. (1)

Sol. $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

for concave mirror

$$\left. \begin{aligned} f &= -ve \\ v &= -ve \\ u &= -ve \end{aligned} \right\}$$



30. The pressure experienced by a swimmer 20 m below the water surface in a lake is approximately [Given density of water = 10^3 kg m^{-3} , $g = 10 \text{ m s}^{-2}$ and $1 \text{ atm} \approx 10^5 \text{ Pa}$]:

- (1) 2 atm
- (2) 3 atm
- (3) 4 atm
- (4) 1 atm

Ans. (2)

Sol. $P = P_{\text{atm}} + \rho gh$
 $= 10^5 + 10^3(10)(20)$
 $= 3 \times 10^5 \text{ Pa}$
 $= 3 \text{ atm}$

31. Match **List-I** with **List-II**. (The symbols carry their usual meaning.)

List-I

A. $\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$

B. $\oint \vec{B} \cdot d\vec{A} = 0$

C. $\oint \vec{E} \cdot d\vec{l} = \frac{-d(\phi)}{dt}$

D. $\oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d(\phi_E)}{dt}$

List-II

I. Ampere-Maxwell's Law

II. Faraday's Law

III. Gauss Law of electrostatics

IV. Gauss Law of magnetism

Choose the **correct** answer from the options given below :

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-IV, B-I, C-III, D-II
- (4) A-III, B-IV, C-II, D-I

Ans. (4)

Sol. Gauss Theorem in electrostatics $\oint \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$

Gauss Theorem in magnetism $\oint \vec{B} \cdot d\vec{s} = 0$

Faraday Law $\oint \vec{E} \cdot d\vec{l} = \frac{-d\phi}{dt}$

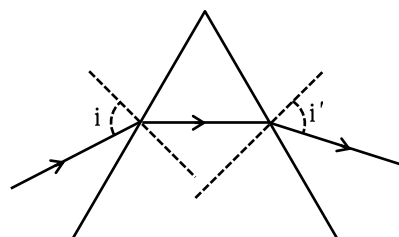
Ampere-Maxwell's Law $\oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d(\phi_E)}{dt}$

32. If i is the angle of incidence and i' is the angle of emergence on a prism, then the relation between i and i' to obtain an angle of minimum deviation is :

- (1) $i > i'$
- (2) $i < i'$
- (3) $i = 0$
- (4) $i = i'$

Ans. (4)

Sol.



For minimum deviation condition :
 $i = i'$



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33. Match List-I with List-II.

List-I (Series)	List-II (Wave number in cm^{-1})
A. Balmer series	I. $R\left(\frac{1}{1^2} - \frac{1}{n^2}\right)$
B. Lyman series	II. $R\left(\frac{1}{4^2} - \frac{1}{n^2}\right)$
C. Brackett series	III. $R\left(\frac{1}{5^2} - \frac{1}{n^2}\right)$
D. Pfund series	IV. $R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$

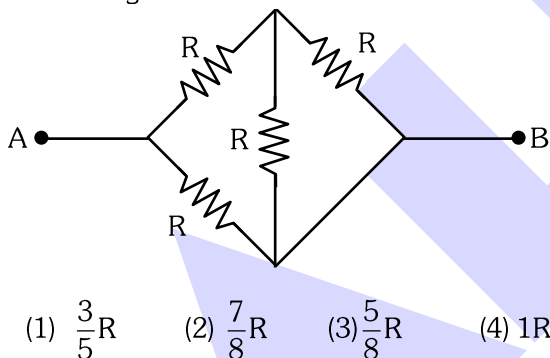
Choose the **correct** answer from the options given below :

- (1) A-II, B-III, C-IV, D-I (2) A-IV, B-I, C-II, D-III
(3) A-III, B-II, C-I, D-IV (4) A-I, B-IV, C-III, D-II

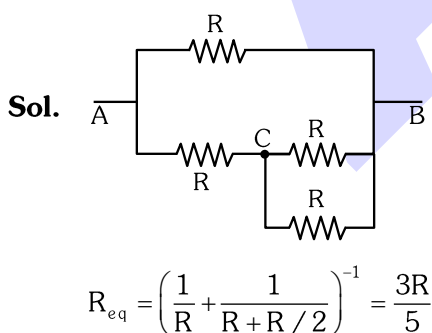
Ans. (2)

- Sol. (A) Balmer series $n \rightarrow 2$ $A \rightarrow \text{IV}$
(B) Lyman series $n \rightarrow 1$ $B \rightarrow \text{I}$
(C) Brackett series $n \rightarrow 4$ $C \rightarrow \text{II}$
(D) Pfund series $n \rightarrow 5$ $D \rightarrow \text{III}$

34. The equivalent resistance R_{AB} between points A and B in the given network is :



Ans. (1)



35. A body weighing 100 N on the surface of earth weighs $x \text{ kg ms}^{-2}$ at a height $\frac{1}{9}R_E$ above the surface of earth. The value of x is ($g = 10 \text{ m s}^{-2}$ at surface of earth and R_E is the radius of earth) :

(1) 54 (2) 81 (3) 62 (4) 72

Ans. (2)

Sol. $W = mg'$

$$= m \left[g \frac{R^2}{(R+h)^2} \right] = mg \left[\frac{R^2}{\left(R + \frac{R}{9} \right)^2} \right]$$

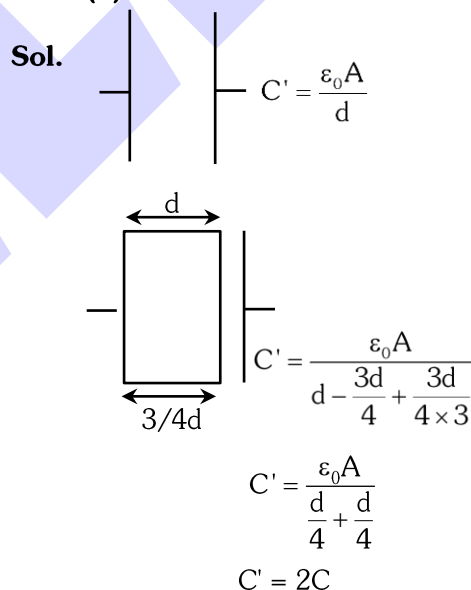
$$= mg \left[\frac{81}{100} \right] = mg \left[\frac{81}{100} \right] = 81N$$

Physics : Section-B (Q. No. 36 to 50)

36. A dielectric slab of dielectric constant 3 having the same area of cross-section as that of parallel plate capacitor but of thickness $\frac{3^{\text{th}}}{4}$ of the separation of plates is inserted into the capacitor. The ratio of potential difference across the plates without dielectric to that with dielectric is :

(1) 2 : 3 (2) 3 : 2 (3) 2 : 1 (4) 1 : 2

Ans. (3)



Now

$$\therefore V = \frac{Q}{C}$$

$$\therefore \frac{V_1}{V_2} = \frac{Q}{C_1} \times \frac{C_2}{Q} = \frac{2}{1}$$



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37. For a smoothly running analog clock, the ratio of number of rotations made in a day by the hour hand and second hand respectively, is :

- (1) 1 : 720 (2) 1 : 60
(3) 2 : 5 (4) 24 : 1

Ans. (1)

Sol. $\frac{\theta_1}{\theta_2} = \frac{T_2}{T_1} = \frac{60}{12 \times 60 \times 60} = \frac{1}{720}$

38. An ideal gas at 0°C and atmospheric pressure P has volume V. The percentage increase in its temperature needed to expand it to 3V at constant pressure is :

- (1) 200 % (2) 300% (3) 50% (4) 100%

Ans. (1)

Sol. $\frac{T_2}{T_1} = \frac{V_2}{V_1} \Rightarrow T_2 = 3T_1$

$\Rightarrow \frac{\Delta T}{T} \times 100\% = \frac{3T_1 - T_1}{T_1} \times 100\% = 200\%$

39. Object A has half the kinetic energy as that of object B. The object B has half the mass as that of the object A. The object A speeds up by 1 ms⁻¹ and then has the same kinetic energy as that of object B. The initial speed of object A is nearly (Take $\sqrt{2} \approx 1.4$) :

- (1) 1 m s⁻¹ (2) 2.5 m s⁻¹
(3) 4.8 m s⁻¹ (4) 0.5 m s⁻¹

Ans. (2)

Sol. $KE_A = \frac{1}{2} KE_B$

$\frac{1}{2} m_A v_A^2 = \frac{1}{2} \left(\frac{1}{2} m_B v_B^2 \right)$

$2v_A^2 = \frac{m_B}{m_A} v_B^2$

Now : $KE'_A = KE_B$

$\frac{1}{2} m_A (v_A + 1)^2 = \frac{1}{2} m_B v_B^2$

$(v_A + 1)^2 = \frac{m_B}{m_A} v_B^2$

$\Rightarrow 2v_A^2 = (v_A + 1)^2$

$\Rightarrow \sqrt{2}v_A = v_A + 1$

$v_A = \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1$

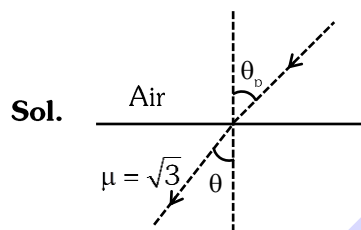
$= 1.4 + 1$

$= 2.4 \text{ m/s} \approx 2.5 \text{ m/s}$

40. If an unpolarised light is incident on a plane surface of refractive index $\sqrt{3}$ at Brewster's angle then the angle of refraction is :

- (1) 30° (2) 60° (3) 90° (4) 0°

Ans. (1)



Sol.

U.P.L. $\mu = \tan \theta_p$
 $\sqrt{3} = \tan \theta_p$
 $\therefore \theta_p = 60^\circ$

Apply Snell's Law

$1 \times \sin \theta_p = \sqrt{3} \sin \theta$

$\theta_p = 60^\circ$

$1 \times \sin 60 = \sqrt{3} \sin \theta$

$\frac{\sqrt{3}}{2} = \sqrt{3} \sin \theta$

$\sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$

41. A body is falling freely in a resistive medium. The motion of the body is described by $\frac{dv}{dt} = 4 - 2v$,

where v is the velocity of body at any instant (in ms⁻¹). The initial acceleration and terminal velocity of the body, respectively, are :

- (1) 2 m s⁻², 4 m s⁻¹ (2) 6 m s⁻², 2 m s⁻¹
(3) 2 m s⁻², 6 m s⁻¹ (4) 4 m s⁻², 2 m s⁻¹

Ans. (4)

Sol. Initial Acceleration (At t = 0 and v = 0)

$a = 4 \text{ m/s}^2$

Terminal velocity (when a = 0)

$\Rightarrow 0 = 4 - 2v \Rightarrow v = 2 \text{ m/s}$

42. A particle of mass m and charge q is placed in a uniform electric field E at t = 0 s. The kinetic energy of the particle after time t is :

- (1) $\frac{E^2 q^2 t^2}{2m}$ (2) $\frac{2E^2 t^2}{qm}$
(3) $\frac{Eq^2 m}{2t^2}$ (4) $\frac{Eqm}{t}$

Ans. (1)



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Sol. $\Delta KE = W$

$$KE - 0 = q\Delta V$$

$$= qEd$$

$$= qE\left(\frac{1}{2}at^2\right)$$

$$= \frac{1}{2}qE\left(\frac{qE}{m}\right)t^2 = \frac{q^2E^2t^2}{2m}$$

43. A rocket is fired vertically upward with a speed of $\frac{v_e}{\sqrt{2}}$ from the earth's surface, where v_e is escape

velocity on the surface of earth. The distance from the surface of earth upto which the rocket can go before returning to the earth is (Given radius of earth = 6400 km) :

- (1) 3200 km
- (2) 6400 km
- (3) 12800 km
- (4) 1600 km

Ans. (2)

Sol. If $v = k v_e$

$$h = \frac{Rk^2}{1-k^2}$$

$$\Rightarrow h = \frac{R\left(\frac{1}{\sqrt{2}}\right)^2}{1-\left(\frac{1}{\sqrt{2}}\right)^2} = R$$

$$= R = 6400 \text{ km}$$

44. A rod of length L rotates with a small uniform angular velocity ω about its perpendicular bisector. A uniform magnetic field B exists parallel to the axis of rotation. The potential difference between the center of the rod and an end is :

- (1) $\frac{B\omega L^2}{2}$
- (2) $\frac{B\omega L^2}{4}$
- (3) zero
- (4) $\frac{B\omega L^2}{8}$

Ans. (4)

Sol. $\text{emf} = \frac{B\omega(L/2)^2}{2} = \frac{B\omega L^2}{8}$

45. If T_1, T_2, T_3, T_4 and T_5 represent the tension in the string of a simple pendulum when the bob is at the left extreme, right extreme, mean, any intermediate left and any intermediate right positions, respectively. Then which of the following relations are correct ?

- (A) $T_1 = T_2$
- (B) $T_3 > T_2$
- (C) $T_4 > T_3$
- (D) $T_3 = T_4$
- (E) $T_5 > T_2$

Choose the **most appropriate answer** from the options given below :

- (1) B, C and D only
- (2) A, B and E only
- (3) C, D and E only
- (4) A, B and C only

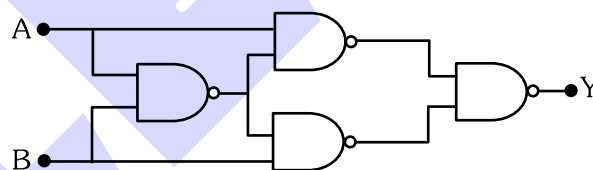
Ans. (2)

Sol. $T_3 > T_4 > T_1 = T_2$

and $T_3 > T_5 > T_1 = T_2$

Because tension decreases as bob moves away from mean position

46. The correct input (A, B)-output(Y) combination for the given circuit is :



- (1) $A = 0, B = 1, Y = 1$
- (2) $A = 1, B = 0, Y = 0$
- (3) $A = 0, B = 0, Y = 1$
- (4) $A = 1, B = 1, Y = 1$

Ans. (1)

Sol. $Y = \overline{\overline{A.B.A}} \cdot \overline{\overline{A.B.B}}$

$$= \overline{\overline{A.B.A}} + \overline{\overline{A.B.B}}$$

$$= (\overline{A.B}).A + (\overline{A.B}).B$$

$$= (\overline{A} + \overline{B}).A + (\overline{A} + \overline{B}).B$$

$$= \overline{A}.A + A.\overline{B} + \overline{A}.B + \overline{B}.B$$

$$= A.\overline{B} + B.\overline{A}$$

XOR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



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47. Arrange the following in the order of their resistance

- (A) (0 to 1A) ranged ammeter.
 (B) (0 to 100 mA) ranged milli-ammeter.
 (C) (0 to 500 μ A) ranged micro-ammeter.
 (D) (0 to 100 V) ranged voltmeter.

Choose the **correct** answer from the options given below :

- (1) $D > C > B > A$
 (2) $D > A > B > C$
 (3) $C > B > A > D$
 (4) $A > B > C > D$

Ans. (1)

Sol. In (A) as Range $\uparrow R_{\text{Ammeter}} \downarrow$

(V) has high resistance then (G) & (A)

48. The ratio of nuclear densities and nuclear volumes of ${}^{56}_{26}\text{Fe}$ and ${}^4_2\text{He}$ are, respectively :

- (1) 14 : 1 and 1 : 1
 (2) 1 : 1 and 14 : 1
 (3) 1 : 1 and 13 : 1
 (4) 13 : 1 and 14 : 1

Ans. (2)

Sol. Ratio of densities of nucleus = 1 : 1

$$\therefore V \propto A$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{A_1}{A_2}$$

$$\frac{V_1}{V_2} = \frac{56}{4} = \frac{14}{1}$$

$$\Rightarrow V_1 : V_2 = 14 : 1$$

49. The length of a magnetized iron bar is L and its magnetic moment is M. When this bar is bent to form a semicircle its magnetic moment is :

- (1) $\frac{M\pi}{2}$
 (2) $\frac{M}{2\pi}$
 (3) $\frac{2M}{\pi}$
 (4) M

Ans. (3)

Sol.

$$m\ell = M$$

$$M' = m2r$$



$$\therefore r = \frac{\ell}{\pi}$$

$$M = \frac{m2\ell}{\pi}$$

$$M = \frac{2M}{\pi}$$

50. There are two heaters A and B. Heater A takes time t_1 to boil a given quantity of water, while B takes time t_2 to boil the same quantity of water across same supply voltage. If the two heaters are connected in series, time taken by this combination to boil the same quantity of water will be :

- (1) $t_1 + t_2$
 (2) $\frac{1}{2}(t_1 + t_2)$
 (3) $\frac{t_1 t_2}{2(t_1 + t_2)}$
 (4) $\frac{t_1 t_2}{t_1 + t_2}$

Ans. (1)

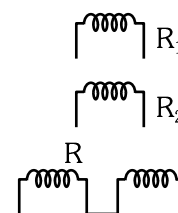
Sol. $Q = P_1 t_1 = V^2 \frac{t_1}{R_1}$

$$Q = P_2 t_2 = V^2 \frac{t_2}{R_2}$$

$$Q = Pt$$

$$Q = \frac{V^2 t}{R_1 + R_2}$$

$$\Rightarrow t = t_1 + t_2$$



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NEET(UG)–2024 (MADHOPUR EXAMINATION)

(Held On Sunday 5th MAY, 2024)

CHEMISTRY

TEST PAPER WITH ANSWER AND SOLUTION

Chemistry : Section-A (Q. No. 51 to 85)

51. Match **List-I** with **List-II** :

List-I (Process/Property)	List-II (Characteristics)
A. Adiabatic process	I. Independent of amount of substance
B. Reversible process	II. Both way process reactant to product and vice-versa
C. Intensive property	III. No transfer of heat between system and surroundings
D. Extensive property	IV. Dependent on amount of substance

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-IV, D-III (2) A-III, B-II, C-I, D-IV
(3) A-I, B-II, C-III, D-IV (4) A-II, B-III, C-IV, D-I

Ans. (2)

- Sol.** (A) Adiabatic $\Rightarrow$ No transfer of heat between system process and surroundings (III)
(B) Reversible $\Rightarrow$ Both way process reactant to product process and vice-versa (II)
(C) Intensive $\Rightarrow$ Independent of property amount of substance (I)
(D) Extensive $\Rightarrow$ Dependent on property amount of substance (IV)

52. Which one of the following molecules is paramagnetic ?

- (1) Li_2 (2) C_2
(3) O_2 (4) H_2

Ans. (3)

Sol. O_2 is paramagnetic due to 2 unpaired e^- in π^*2p_x and π^*2p_y orbitals.

53. 4.74 g of an inorganic compound contains 0.39 g of K, 0.27 g of Al, 1.92 g of SO_4 radicals and 2.16 g of water. If molar mass of the compound is 948 g mol^{-1} , the molecular formula of the inorganic compound is :

- (1) $\text{K}_2\text{Al}_2(\text{SO}_4)_4 \cdot 12 \text{ H}_2\text{O}$
(2) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24 \text{ H}_2\text{O}$
(3) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 12 \text{ H}_2\text{O}$
(4) $\text{KAl}(\text{SO}_4)_2 \cdot 12 \text{ H}_2\text{O}$

Ans. (2)

	K	Al	SO_4	H_2O
Mass : –	0.39	0.27	1.92	2.16
Sol. Mole : –	0.01	0.01	0.02	0.12
Ratio : –	1	1	2	12
	2	2	4	24

54. Given below are two statements :

Statement I : Benzoic acid produces effervescence on treatment with aq. NaHCO_3 .

Statement II : The effervescence is due to the release of hydrogen gas.

In the light of the above statement, choose the **most appropriate** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct.
(4) Both **Statement I** and **Statement II** are correct.

Ans. (2)

Sol. $\text{Ph} - \text{COOH} + \text{NaHCO}_3 \rightleftharpoons \text{PhCOO}^\ominus\text{Na}^\oplus + \text{H}_2\text{CO}_3 (\text{CO}_2 \uparrow)$



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55. Which of the following pair of aqueous solutions will have the same value of osmotic pressure ?

(Assume complete dissociation in aqueous solution)

- (1) 0.1 M Na_3PO_4 and 0.1 M K_2SO_4
 (2) 0.2 M NaCl and 0.1 M K_2SO_4
 (3) 0.2 M NaCl and 0.1 M $\text{K}_3[\text{Fe}(\text{CN})_6]$
 (4) 0.1 M BaCl_2 and 0.2 M K_2SO_4

Ans. (3)

Sol. $\pi = iCRT$

$$\pi_1 = \pi_2 \text{ when } i_1 C_1 = i_2 C_2$$

check options,

56. Given below are two statements :

Statement I : 2F electricity is required for the oxidation of 1 mole H_2O to O_2 .

Statement II : To get 40.0 g of Aluminium from molten Al_2O_3 required electricity is 4.44 F.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both **Statement I** and **Statement II** are false.
 (2) **Statement I** is true but **Statement II** is false.
 (3) **Statement I** is false but **Statement II** is true.
 (4) Both **Statement I** and **Statement II** are true.

Ans. (4)

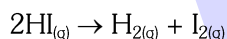
Sol. St.-1 :- 1 mole $\text{H}_2\text{O} \xrightarrow{(-2)} \text{O}_2^{(0)}$

V.F of $\text{H}_2\text{O} = 2$

$$\text{geq. of } \text{H}_2\text{O} = 1 \text{ mol} \times (\text{V.F} = 2) = 2\text{F}$$

$$\text{St.-2 :- geq. of Al} = \frac{w}{E} = \frac{40}{(27/3)} = 4.44 = 4.44 \text{ F}$$

57. Which of the following expression is correct for the reaction given below ?



- (1) $\frac{-\Delta[\text{HI}]}{\Delta t} = \frac{4\Delta[\text{I}_2]}{\Delta t}$ (2) $\frac{-\Delta[\text{HI}]}{\Delta t} = \frac{4\Delta[\text{H}_2]}{\Delta t}$
 (3) $\frac{-\Delta[\text{HI}]}{\Delta t} = \frac{\Delta[\text{H}_2]}{\Delta t}$ (4) $\frac{-\Delta[\text{HI}]}{\Delta t} = \frac{2\Delta[\text{H}_2]}{\Delta t}$

Ans. (4)

Sol. $2\text{HI} \rightarrow \text{H}_2 + \text{I}_2$

$$-\frac{1}{2} \times \frac{\Delta[\text{HI}]}{\Delta t} = \frac{\Delta[\text{H}_2]}{\Delta t}$$

$$\frac{-\Delta[\text{HI}]}{\Delta t} = 2 \times \frac{\Delta[\text{H}_2]}{\Delta t}$$

58. The correct 'spin only' magnetic moments of $[\text{CoF}_6]^{3-}$ and $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ respectively, are :

- (1) 0 BM and 0 BM (2) 4.9 BM and 0 BM
 (3) 4.9 BM and 4.9 BM (4) 0 BM and 4.9 BM

Ans. (2)

59. Match **List-I** with **List-II** :

List-I (Element)	List-II (Most Common oxidation state/s)
A. Fe	I. +2, +7
B. V	II. +3, +2
C. Mn	III. +4
D. Ti	IV. +5

Choose the correct answer from the options given below :

- (1) A-IV, B-II, C-I, D-III
 (2) A-II, B-I, C-IV, D-III
 (3) A-I, B-IV, C-II, D-III
 (4) A-II, B-IV, C-I, D-III

Ans. (4)

Sol. $\text{Fe} = +2, +3, \text{V} \Rightarrow +5, \text{Mn} \Rightarrow +2, +7, \text{Ti} \Rightarrow +4$

60. The technique used to purify liquids having high boiling points and getting decomposed at or below their boiling points is :

- (1) steam distillation
 (2) simple distillation
 (3) distillation under reduced pressure
 (4) fractional distillation

Ans. (3)

61. The method used for quantitative estimation of halogens is :

- (1) Carius method
 (2) Kjeldahl's method
 (3) Combustion method
 (4) Dumas method

Ans. (1)

62. Effective collisions are known to possess :

- A. energy greater than threshold energy.
 B. breaking of old bond in reactant.
 C. formation of new bond in product.
 D. high activation energy
 E. proper orientation.

Choose the correct answer from the options given below :

- (1) A, B, C, E only (2) A, C, D, E only
 (3) B, C, D, E only (4) A, B, C, D only

Ans. (1)

Sol. Fact



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63. Match List-I with List-II :

List-I (complex ion)	List-II (Hybridisation)
A. $[\text{Ni}(\text{CN})_4]^{2-}$	I. sp^3
B. $[\text{CoF}_6]^{3-}$	II. d^2sp^3
C. $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$	IV. sp^3d^2
D. $[\text{NiCl}_4]^{2-}$	V. dsp^2

Choose the correct answer from the options given below :

- (1) A-III, B-I, C-IV, D-II
(2) A-I, B-II, C-III, D-IV
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-III, C-II, D-I

Ans. (4)

Sol. $[\text{Ni}(\text{CN})_4]^{2-} \Rightarrow \text{dps}^2$, $[\text{CoF}_6]^{3-} \Rightarrow \text{sp}^3\text{d}^2$
(CN^- is SFL) F^- is WFL
 $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-} \Rightarrow \text{d}^2\text{sp}^3$, $[\text{NiCl}_4]^{2-} \Rightarrow \text{sp}^3$
 $\text{C}_2\text{O}_4^{2-}$ is SFL with Co^{+3} , Cl^- is WFL

64. Which of the following reaction is **not** a disproportionation reaction ?

- (1) $\text{Cl}_{2(\text{g})} + 2\text{OH}^-_{(\text{aq})} \rightarrow \text{ClO}^-_{(\text{aq})} + \text{Cl}^-_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$
(2) $2\text{NO}_{2(\text{g})} + 2\text{OH}^-_{(\text{aq})} \rightarrow \text{NO}_{2(\text{aq})}^- + \text{NO}_{3(\text{aq})}^- + \text{H}_2\text{O}_{(\text{l})}$
(3) $2\text{H}_2\text{O}_{2(\text{aq})} \rightarrow 2\text{H}_2\text{O}_{(\text{l})} + \text{O}_{2(\text{g})}$
(4) $2\text{F}_{2(\text{g})} + 2\text{OH}^-_{(\text{aq})} \rightarrow 2\text{F}^-_{(\text{aq})} + \text{OF}_{2(\text{g})} + \text{H}_2\text{O}_{(\text{l})}$

Ans. (4)

Sol. Fact

65. Match List-I with List-II :

List-I (IUPAC official name)	List-II (IUPAC symbol)
A. Mendelevium	I. Mt
B. Meitnerium	II. Mc
C. Moscovium	III. No
D. Nobelium	IV. Md

Choose the correct answer from the options given below :

- (1) A-IV, B-II, C-I, D-III
(2) A-IV, B-I, C-II, D-III
(3) A-I, B-IV, C-II, D-III
(4) A-III, B-I, C-II, D-IV

Ans. (2)

66. CCl_2F_2 is one of the most common haloalkanes used in industries. The reaction through which it is manufactured from tetrachloromethane is called :

- (1) Sandmeyer's reaction
(2) Swarts reaction
(3) Wurtz-Fittig reaction
(4) Finkelstein reaction

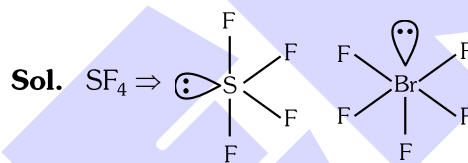
Ans. (2)

Sol. Freon (CCl_2F_2) is manufactured from CCl_4 by Swarts reaction.

67. Which pair of the following compounds has one lone pair of electrons on the central atom ?

- (1) SF_4 , ClF_3 (2) ClF_3 , XeF_4
(3) XeF_4 , BrF_5 (4) SF_4 , BrF_5

Ans. (4)



68. Match List-I with List-II :

List-I (Reactions)	List-II (Products)
A. $\text{CH}_3\text{CH}=\text{CH}_2 + \text{H}_2\text{O} + \text{O} \xrightarrow[273\text{K}]{\text{dil. KMnO}_4}$	I. $(\text{CH}_3)_2\text{C}=\text{O} + \text{CO}_2 + \text{H}_2\text{O}$
B. $\text{CH}_3\text{CH}=\text{CH}_2 \xrightarrow[\text{(ii) Zn+H}_2\text{O}]{\text{(i) O}_3}$	II. $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{OH}$
C. $\text{H}_3\text{C}-\text{C}(\text{H})=\text{CH}_2 + \text{H}_2\text{O} \xrightarrow{\text{H}^+}$	III. $\text{HCHO} + \text{CH}_3-\text{C}(=\text{O})-\text{H}$
D. $\text{H}_3\text{C}-\text{C}(\text{H})=\text{CH}_2 \xrightarrow{\text{KMnO}_4/\text{H}^+}$	IV. $(\text{CH}_3)_3\text{C}-\text{OH}$

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
(2) A-II, B-III, C-I, D-IV
(3) A-IV, B-I, C-III, D-II
(4) A-II, B-IV, C-III, D-I

Ans. (1)

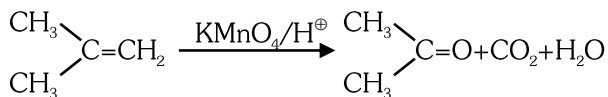
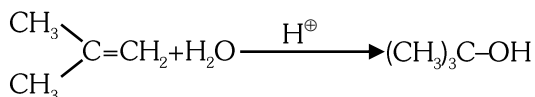
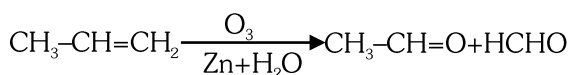
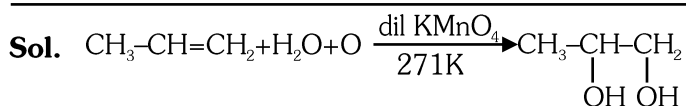


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69. The behaviour of samarium ($Z=62$) is very much like europium ($Z=63$). Since both elements :

- (1) have $4f^5 6s^2$ and $4f^6 6s^2$ electronic configurations, respectively.
- (2) have $4f^7 6s^2$ and $4f^8 6s^2$ electronic configurations, respectively.
- (3) have same atomic radii.
- (4) exhibit +2 and +3 oxidation states.

Ans. (4)

Sol. NCERT, Page No. # 109 Last line

70. Match **List-I** with **List-II** :

List-I (Vitamins)	List-II (Deficiency / other name / property)
A. Vitamin K	I. RBC deficient in haemoglobin
B. Vitamin B ₁₂	II. Bleeding gums
C. Vitamin C	III. Thiamine
D. Vitamin B ₁	IV. Fat soluble

Choose the **correct** answer from the options given below :

- (1) A-IV, B-III, C-II, D-I
- (2) A-I, B-IV, C-II, D-III
- (3) A-IV, B-I, C-II, D-III
- (4) A-I, B-II, C-III, D-IV

Ans. (3)

- Sol.** (A) Vitamin K → IV (fat soluble)
 (B) Vitamin B₁₂ → I (RBC deficient in Haemoglobin)
 (C) Vitamin C → II (Bleeding gums)
 (D) Vitamin B₁ → III (Thiamine)

71. Match **List-I** with **List-II** :

List-I (Test)	List-II (Compound identified)
A. Carbylamine test	I. phenol
B. Bayer's test	II. acetone
C. Iodoform test	III. ethylene
D. Phthalein dye test	IV. aniline

Choose the **correct** answer from the options given below :

- (1) A-IV, B-II, C-III, D-I
- (2) A-IV, B-II, C-I, D-III
- (3) A-II, B-III, C-I, D-IV
- (4) A-IV, B-III, C-II, D-I

Ans. (4)

- Sol.** (A) Carbyl amine test (IV) Aniline
 (B) Bayer's test (III) Ethylene
 (C) Iodoform test (II) Acetone
 (D) Phthalein dye test (I) Phenol

72. Molar mass of a compound (X) whose 2.6 mol weighs 312 g is :

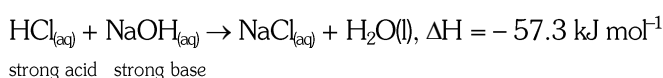
- (1) 120 g mol⁻¹
- (2) 60 g mol⁻¹
- (3) 811.2 g mol⁻¹
- (4) 312 g mol⁻¹

Ans. (1)

Sol. Moles = $\frac{\text{Mass}}{\text{Mol. mass}}$

$\Rightarrow \text{Mol. mass} = \frac{312}{2.6} = 120 \text{ g mol}^{-1}$

73. The standard enthalpy of neutralization is $-57.3 \text{ kJ mol}^{-1}$ as :



The enthalpy of neutralization of 0.25 mole of HCl by 0.25 mol of NaOH is :

- (1) $-28.3 \text{ kJ mol}^{-1}$
- (2) $-14.32 \text{ kJ mol}^{-1}$
- (3) $+57.3 \text{ kJ mol}^{-1}$
- (4) $-57.3 \text{ kJ mol}^{-1}$

Ans. (2)

- Sol.** For 1 mole HCl and NaOH; Heat = -57.3 kJ
 For 0.25 mole; Heat = -57.3×0.25
 $= -14.32 \text{ kJ}$
 (Unit should not be kJ mol^{-1})



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74. Match List-I with List-II

List-I (Amine)	List-II (pK _b value)
A. Benzenamine	I. 3.38
B. Phenylmethanamine	II. 9.38
C. Methanamine	III. 4.70
D. N-Ethylethanamine	IV. 3.0

Choose the **correct** answer from the options given below :

- (1) A-III, B-II, C-I, D-IV (2) A-II, B-I, C-III, D-IV
(3) A-I, B-III, C-II, D-IV (4) A-II, B-III, C-I, D-IV

Ans. (4)

Sol. **Basic strength order** $\propto \frac{1}{pK_b}$:-

N-Ethyl ethane anine > Methanamine > Phenyl > Benzenamine
(3.0) (3.38) methyl amine (9.38)
(4.70)

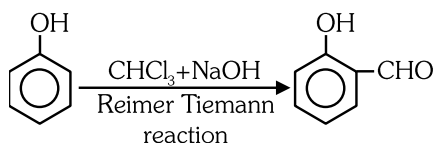
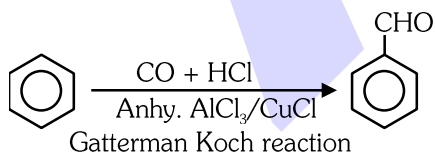
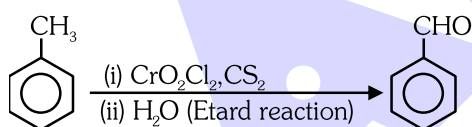
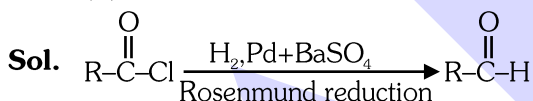
75. Match List-I with List-II

List-I (Reagent)	List-II (Name of the reaction)
A. H ₂ , Pd-BaSO ₄	I. Gattermann-Koch reaction
B. (i) CrO ₂ Cl ₂ , CS ₂ ; (ii) H ₂ O	II. Reimer-Tiemann reaction
C. CO, HCl, Anhyd. AlCl ₃ /CuCl	III. Etard reaction
D. CHCl ₃ , NaOH	IV. Rosenmund reduction

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II (2) A-I, B-III, C-IV, D-II
(3) A-II, B-I, C-III, D-IV (4) A-II, B-III, C-I, D-IV

Ans. (1)



76. The common oxidation state shown by actinoids is +3 with the other oxidation states being +4, +5, +6 and +7. The number of oxidation states shown by the following elements decrease in the order :

- A. Np (Z = 93) B. Am (Z = 95)
C. Ac (Z = 89) D. Pa (Z = 91)

Choose the **correct** answer from the options given below :

- (1) A > B > C > D (2) B > A > D > C
(3) D > B > A > C (4) A > B > D > C

Ans. (4)

Sol. Np - +7 Highest oxidation state
Am - +6 Highest oxidation state
Pa - +5 Highest oxidation state
Ac - +3 Highest oxidation state

77. Match List-I with List-II :

List-I Equilibria	List-II Conditions/Parameters
A. Liquid $\rightleftharpoons$ Vapour	I. Melting point
B. Solid $\rightleftharpoons$ Liquid	II. Boiling point
C. Solid $\rightleftharpoons$ Vapour	III. Sublimation point
D. Solute (Solid) $\rightleftharpoons$ Solute (solution)	IV. Saturated solution V. Unsaturated solution

Choose the **correct** answer from the options given below :

- (1) A-I, B-II, C-III, D-IV (2) A-III, B-II, C-I, D-IV
(3) A-II, B-I, C-III, D-V (4) A-II, B-I, C-III, D-IV

Ans. (4)

Sol. (A) $\rightarrow$ (III), (B) $\rightarrow$ (I), C $\rightarrow$ (III), (D) $\rightarrow$ (IV)

78. Arrange the following hydrides in the order of decreasing bond angle :

- A. SbH₃
B. AsH₃
C. PH₃
D. NH₃

Choose the **correct** answer from the options given below :

- (1) B > A > C > D (2) D > C > B > A
(3) A > C > B > D (4) B > A > D > C

Ans. (2)

Sol. Order of bond angle
 $\Rightarrow \text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$
(Bond angle $\propto$ EN of central atom)



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79. Consider the reaction in equilibrium

$\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ at 500 K. The concentration of $\text{PCl}_5 = 1.40 \text{ M}$ concentration of $\text{Cl}_2 = 1.60 \text{ M}$, concentration of $\text{PCl}_3 = 1.60 \text{ M}$. Calculate K_c .

- (1) 2.6 (2) 1.83
(3) 3.4 (4) 2.00

Ans. (2)

Sol. $K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{1.6 \times 1.6}{1.4} = 1.83$

80. Isotope of an element contains 19.23% more neutrons as compared to protons. The correct element along with its mass number is :

(Given Atomic number Fe : 26, Co : 27)

- (1) ^{57}Fe
(2) ^{57}Co
(3) ^{60}Co
(4) ^{56}Fe

Ans. (1)

Sol. No. of protons = p

No. of neutrons = $p + \frac{19.23}{100} \times p$

Option check ; $n + p = 2.1923p$

$p = 26$; $n + p = 57$

81. Given below are two statements:

Statement-I : The energy of He^+ ion in $n = 2$ state is same as the energy of H atom in $n = 1$ state.

Statement-II : It is possible to determine simultaneously the exact position and exact momentum of an electron in H atom.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are false.
(2) **Statement I** is true but **Statement II** is false.
(3) **Statement I** is false but **Statement II** is true.
(4) Both **Statement I** and **Statement II** are true.

Ans. (2)

Sol. St.-1 $\therefore E = -13.6 \times \frac{z^2}{n^2}$

For H-atom in $n = 1$; $E = -13.6 \text{ eV}$

For He^+ in $n = 2$, $E = -13.6 \times \frac{(2)^2}{(2)^2}$
 $= -13.6 \text{ eV}$

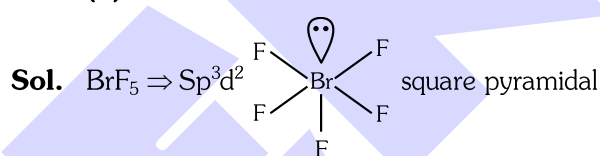
St.1 is correct

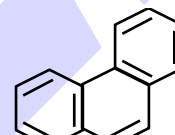
St.-2 $\therefore$ Incorrect (uncertainty principle)

82. The **correct** shape and hybridisation of BrF_5 is :

- (1) Square pyramidal and d^2sp^3
(2) Trigonal bipyramidal and sp^3d^2
(3) Trigonal bipyramidal and d^2sp^3
(4) Square pyramidal and sp^3d^2

Ans. (4)



83. Phenanthrene, , is an

aromatic compound and follows Huckel's $(4n + 2)\pi$ electron rule. The value of n is

- (1) 1 (2) 2 (3) 3 (4) 0

Ans. (3)

Sol. $14\pi e^- = (4n + 2)$

$n = 3$

84. Identify the **incorrect** statement :

- (1) Na_2O is a basic oxide and Cl_2O_7 is an acidic oxide.
(2) The following four species are called isoelectronic species : O^{2-} , F^- , Na^+ and Mg^{2+} .
(3) Among the four species Mg , Al , Mg^{2+} and Al^{3+} , the smallest one is Al .
(4) The oxidation and coordination number (or covalency) of Al in $[\text{AlCl}(\text{H}_2\text{O})_5]^{2+}$ are +3 and 6, respectively

Ans. (3)

Sol. Correct order of radius - $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Al} < \text{Mg}$

Al in $[\text{AlCl}(\text{H}_2\text{O})_5]^{2+}$ are +3 and 6, respectively



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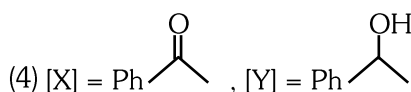
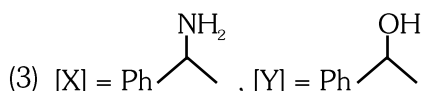
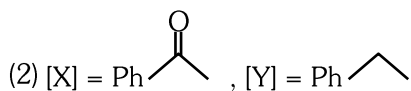
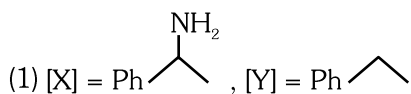
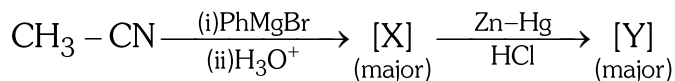
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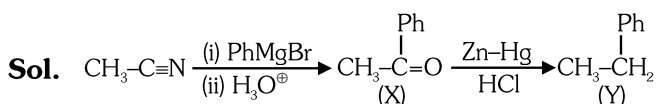
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85. Identify [X] and [Y] in given reaction :

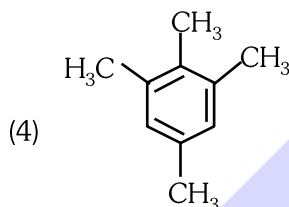
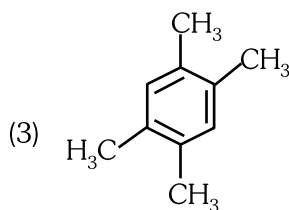
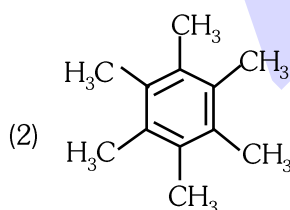
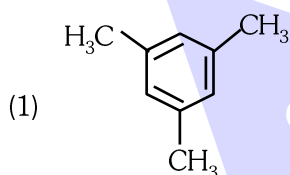
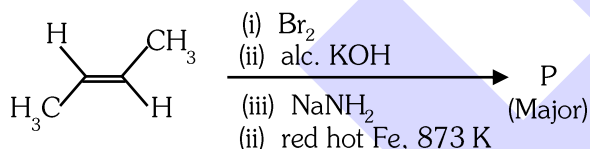


Ans. (2)

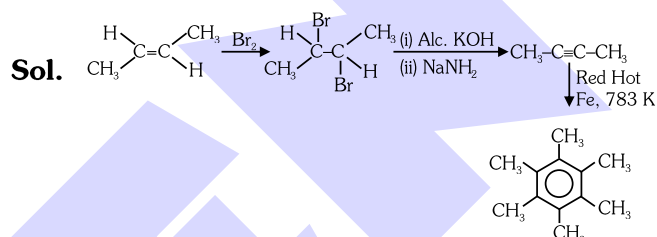


Chemistry : Section-B (Q. No. 86 to 100)

86. The major product P formed in the following reaction sequence is :



Ans. (2)



87. The Crystal Field Stabilisation Energy (CFSE) and the 'spin only' magnetic moment of $[\text{FeF}_6]^{3-}$ ion respectively, are :

- (1) $0 \Delta_0$ and 1.73 BM (2) $2.0 \Delta_0$ and 5.9 BM
(3) $2.0 \Delta_0$ and 1.73 BM (4) $0 \Delta_0$ and 5.9 BM

Ans. (4)

Sol. $[\text{FeF}_6]^{3-} \Rightarrow \text{Fe}^{+3} = 3d^5 (t_{2g}^3 e_g^2) \text{ F}^- \text{ is WFL.}$

$$\text{CFSE} = -0.4 \Delta_0 \times 3 + 0.6 \Delta_0 \times 2$$

$$\text{CFSE} = 0$$

$$\text{Number of unpaired } e^- = 5$$

$$\mu = \sqrt{5(5+2)} = 5.9 \text{ BM.}$$

88. When 5 g of a non-volatile non-electrolyte solute dissolved in 100 g of a certain solvent, the freezing point of the solvent decreases by 0.25 K. The molar mass of the solute is :

$$[K_f \text{ of the given solvent} = 1.2 \text{ K kg mol}^{-1}]$$

- (1) 238.2 g mol^{-1} (2) 241.8 g mol^{-1}
(3) 240.0 g mol^{-1} (4) 242.8 g mol^{-1}

Ans. (3)

$$\Delta T_f = i \times K_f \times \text{molality}$$

$$0.25 = 1 \times 1.2 \times \left[\frac{5/m}{100/1000} \right]$$

$$m = 240$$



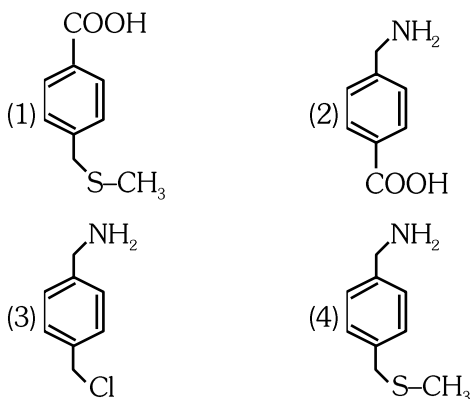
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89. The compound that gives blood red colour in Lassaigne's test is :



Ans. (4)

Sol. Lassaigne's test
Blood Red colour → N and S

90. For the equilibrium
 $2 \text{NOCl}_{(g)} \rightleftharpoons 2 \text{NO}_{(g)} + \text{Cl}_{2(g)}$

the value of equilibrium constant, K_c is 3.0×10^{-6} at 1000 K. Find K_p for the reaction at this temperature.

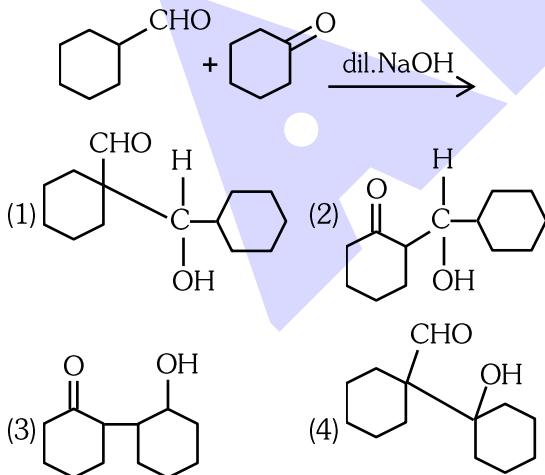
(Given $R : 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

- (1) 2.494×10^{-2} (2) 3.0×10^{-6}
(3) 2.494×10^{-4} (4) 1.493

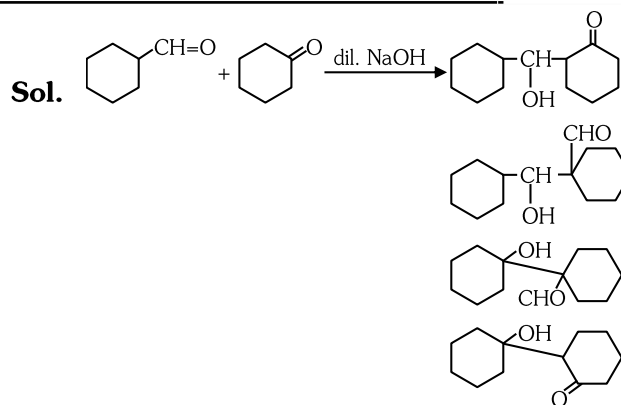
Ans. (3)

Sol. $K_p = K_c \times (RT)^{\Delta n_g}$
 $= 3 \times 10^{-6} \times (0.0831 \times 1000)^{\Delta n_g=1}$
 $= 0.2493 \times 10^{-3}$
 $= 2.49 \times 10^{-4}$

91. The product that **cannot** be formed in the following reaction is :



Ans. (3)



92. Identify the **incorrect** statement :

- (1) Boron trioxide is acidic, aluminium and gallium trioxides are amphoteric while indium and thallium trioxides are basic.
(2) The hybridisation of Al in $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ is d^2sp^3 .
(3) Two isotopes of Boron, namely ^{10}B (19%) and ^{11}B (81%) are known.
(4) The relative stability of group 13 elements in +1 oxidation state varies as $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$.

Ans. (2)

Sol. hybridisation of Al in $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ is sp^3d^2 .

93. In salt analyses, group V radicals (Ba^{2+} , Ca^{2+} and Sr^{2+}) are precipitated as their carbonates by adding solid NH_4Cl and excess of NH_4OH solution followed by solid $(\text{NH}_4)_2\text{CO}_3$.

Choose the correct reagent used for the confirmation of Ca^{2+} ion from the following :

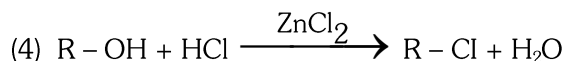
- (1) Potassium chromate (2) Ammonium oxalate
(3) Ammonium nitrate (4) Ammonium sulphate

Ans. (2)

Sol. Ammonium oxalate is used for the confirmation of Ca^{2+} .

94. The most preferred method for the preparation of pure alkyl chlorides from alcohols is :

- (1) $\text{R-OH} + \text{PCl}_5 \rightarrow \text{R-Cl} + \text{POCl}_3 + \text{HCl}$
(2) $\text{R-OH} + \text{SOCl}_2 \rightarrow \text{R-Cl} + \text{SO}_2 + \text{HCl}$
(3) $\text{R-OH} + \text{PCl}_3 \rightarrow \text{R-Cl} + \text{H}_3\text{PO}_3$



Ans. (2)

Sol. $\text{R-OH} + \text{SOCl}_2 \rightarrow \text{R-Cl} + \text{SO}_2 \uparrow + \text{HCl} \uparrow$

By products are in gaseous state, so separation of products not required.



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95. Given below are two statements :

Statement I : Benzenediazonium chloride is a colourless crystalline solid. It is insoluble in water but reacts with water when warmed.

Statement II : Benzenediazonium chloride on reacting with HCl in presence of copper powder gives chlorobenzene as the product. This is an example of Gatterman reaction.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are false
- (2) **Statement II** is true but **Statement I** is false
- (3) **Statement I** is false but **Statement II** is true
- (4) Both **Statement I** and **Statement II** are true

Ans. (3)

96. Number of mole of MnO_4^- getting reduced to Mn^{2+} under acidic condition by 4.517×10^{24} electrons is :

- (1) 7.5 moles
- (2) 2.5 moles
- (3) 5.0 moles
- (4) 1.5 moles

Ans. (4)

Sol. $\text{MnO}_4^- + 5e^- \rightarrow \text{Mn}^{2+}$

$5 \times (6.02 \times 10^{23}) e^- \xrightarrow{\text{reduce}} 1 \text{ mole } \text{MnO}_4^-$

$4.517 \times 10^{24} e^- \rightarrow \frac{1}{(30.1 \times 10^{23})} \times (4.517 \times 10^{24})$
 $= 1.5 \text{ mole}$

97. Time taken by the first order decomposition of SO_2Cl_2 to decompose to 40% is 560 seconds. Rate constant for the reaction is:

($\log 2.5 = 0.3979$)

- (1) $2.276 \times 10^{-5} \text{ min}^{-1}$
- (2) $2.216 \times 10^{-5} \text{ min}^{-1}$
- (3) $2.126 \times 10^{-5} \text{ min}^{-1}$
- (4) $2.726 \times 10^{-5} \text{ min}^{-1}$

Ans. (BONUS)

Sol. $K = \frac{2.303}{\left(t = \frac{560}{60} \text{ min}\right)} \times \log\left(\frac{100}{40} = 2.5\right)$
 $= \frac{2.303 \times 60}{560} \times 0.3979$
 $= 9.81 \times 10^{-2} \text{ min}^{-1}$

98. In salt analyses, Cl^- , Br^- and I^- are identified by adding dilute HNO_3 followed by AgNO_3 solution to the salt solution. Choose the correct statements from the following :

- A. Cl^- gives pale yellow precipitate.
- B. Br^- gives pale yellow precipitate which is partially soluble in NH_4OH .
- C. I^- gives yellow precipitate which is insoluble in NH_4OH .
- D. Cl^- gives white precipitate which is insoluble in NH_4OH .
- E. I^- gives purple violet gas.

Choose the **correct** answer from the options given below :

- (1) A and B only
- (2) A and E only
- (3) D and E only
- (4) B and C only

Ans. (4)

Sol. A - Cl^- gives yellow ppt - false

Reason - it gives white ppt of AgCl

D - Cl^- gives white ppt which is insoluble in NH_4OH - false

Reason - white ppt of AgCl is soluble in NH_4OH due to formation of soluble complex.

99. Identify the **incorrect** statement :

- (1) Silicon and Germanium are useful in semiconductor industry.
- (2) Boron trifluoride is a strong Lewis acid.
- (3) Common oxidation states of group 14 elements are +1 and +3
- (4) Carbon has the tendency to form chains and rings with itself

Ans. (3)

Sol. Common oxidation of group 14 elements are +2 and +4.

100. Enthalpy of combustion of carbon to carbon dioxide is $-390.0 \text{ kJ mol}^{-1}$. The amount of heat released, when 35.0 g of CO_2 is formed from the reaction of carbon and dioxygen gas, is :

- (1) 490 kJ
- (2) 245 kJ
- (3) 700 kJ
- (4) 310 kJ

Ans. (4)

Sol. $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$; $\Delta_c H = -390 \text{ kJ mol}^{-1}$

1 mole = 44 g $\rightarrow 390 \text{ kJ released}$

For 35 g $\rightarrow \frac{390}{44} \times 35 = 310 \text{ kJ}$



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NEET(UG)–2024 (MADHOPUR EXAMINATION)

(Held On Sunday 5th MAY, 2024)

BIOLOGY

TEST PAPER WITH ANSWER AND SOLUTION

Botany : Section-A (Q. No. 101 to 135)

101. Given below are two statements :

Statement I : When the fitness of one species is significantly lower in the presence of another species, the process is defined as competition.

Statement II : When fungi remain in association with living plants or animals they are called Saprophytes.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are false
- (2) **Statement I** is true but **Statement II** is false
- (3) **Statement I** is false but **Statement II** is true
- (4) Both **Statement I** and **Statement II** are true

Ans. (2)

Sol. NCERT-XII, Pg. # 198, 201

102. Which of the following statement is **incorrect** about enzymes ?

- (1) In thermophilic organisms enzymes can catalyze reaction at high temperature i.e. 90°C
- (2) All enzymes are proteinaceous in nature
- (3) Some enzymes have metal ions
- (4) They are highly substrate specific

Ans. (2)

Sol. NCERT-XI, Pg. # 112, 113, 114, 117

103. Important objective of biotechnology in agriculture is to :

- (1) increase nitrogen content in plant.
- (2) decrease the seed number.
- (3) increase the plant weight.
- (4) produce pest resistant varieties of plant.

Ans. (4)

Sol. NCERT-XII, Pg. # 179

104. The negatively geotropic roots produced by *Rhizophora* are called :

- (1) Prop roots
- (2) Stilt roots
- (3) Storage roots
- (4) Pneumatophores

Ans. (4)

Sol. NCERT-XI, Pg. # 67 (Old NCERT)

105. In glycolysis, the conversion of 1, 3-bisphospho Glyceric Acid to 3-Phosphoglyceric Acid is :

- (1) Energy utilising process
- (2) Phosphorylation process
- (3) Isomerisation process
- (4) Energy yielding process

Ans. (4)

Sol. NCERT-XI, Pg. # 156

106. Which of the following is **not** a selectable marker of cloning vectors ?

- (1) Metformin
- (2) Chloramphenicol
- (3) Tetracycline
- (4) Ampicillin

Ans. (1)

Sol. NCERT-XII, Pg. # 169

107. The transverse section of monocot root shows the following internal tissue organization. Arrange them in correct sequence starting from periphery to centre.

- | | |
|---------------|--------------|
| A. Endodermis | B. Pith |
| C. Epidermis | D. Pericycle |
| E. Cortex | |

Choose the **correct** answer from the options given below :

- | | |
|-------------------|-------------------|
| (1) A, C, E, B, D | (2) C, E, A, D, B |
| (3) C, E, D, B, A | (4) D, C, E, A, B |

Ans. (2)

Sol. NCERT-XI, Pg. # 74



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108. Given below are two statement :

Statement I : Two DNA sequences where prepared corresponding to A and B, chains of human insulin and were introduced in the plasmids of *Agrobacterium* to produce insulin chain.

Statement II : Chain A and B were produced separately, extracted and combined by creating disulphide bonds to form human insulin.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are false
- (2) **Statement I** is true but **Statement II** is false
- (3) **Statement I** is false but **Statement II** is true
- (4) Both **Statement I** and **Statement II** are true

Ans. (3)

Sol. NCERT-XII, Pg. # 182

109. ABO blood groups is an example of :

- (1) Multiple alleles
- (2) Pleiotropy
- (3) Linkage
- (4) Incomplete dominance

Ans. (1)

Sol. NCERT-XII, Pg. # 61

110. Match **List-I** with **List-II**.

List-I	List-II
A. Microscopic angiosperm	I. <i>Salvinia</i>
B. Tallest gymnosperm	II. <i>Marchantia</i>
C. Thalloid bryophyte	III. <i>Sequoia</i>
D. Heterosporous pteridophyte	IV. <i>Wolffia</i>

Choose the **correct** answer from the options given below :

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-II, B-I, C-IV, D-III
- (4) A-I, B-III, C-II, D-IV

Ans. (1)

Sol. NCERT-XI, Pg. # 29, 32, 34

111. Given below are two statements :

Statement I : Rain forests which used to cover more than 14% of the earth's land surface is now reduced to 6%.

Statement II : Amazon rain forest has greatest biodiversity on earth.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both **Statement I** and **Statement II** are false
- (2) **Statement I** is true but **Statement II** is false
- (3) **Statement I** is false but **Statement II** is true
- (4) Both **Statement I** and **Statement II** are true

Ans. (4)

Sol. NCERT-XII, Pg. # 222

112. Synthesis of ATP linked to development of a proton gradient across a membrane is :

- (1) Wobble hypothesis
- (2) Chemiosmotic hypothesis
- (3) Rivet Popper hypothesis
- (4) Mass flow hypothesis

Ans. (2)

Sol. NCERT-XI, Pg. # 140

113. The given figure with reference to the anatomy of plants represents :



- (1) Xylem fibre
- (2) Xylem parenchyma
- (3) Vessel
- (4) Tracheid

Ans. (3)

Sol. NCERT-XI, Pg. # 87 (Old NCERT)



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114. The population growing in a habitat with limited resources will show :

- A. lag phase, followed by phases of acceleration and deceleration and finally an asymptote.
- B. the ability to realise its innate potential to grow in number and reach enormous densities in short time.
- C. exponential growth.
- D. logistic growth.

Choose the correct answer from the options given below :

- (1) C and D only (2) A and D only
- (3) B and C only (4) A and B only

Ans. (2)

Sol. NCERT-XII, Pg. # 194, 195

115. The following can be found as a zwitter ion :

- (1) Monosaccharide (2) Amino acid
- (3) Nucleic acid (4) Fatty acid

Ans. (2)

Sol. NCERT-XI, Pg. # 106

116. The phenomenon where a single gene exhibiting multiple phenotypic expressions is called :

- (1) Pleiotropy (2) Multiple alleles
- (3) Polygenic inheritance (4) Co-dominance

Ans. (1)

Sol. NCERT-XII, Pg. # 69

117. Match **List-I** with **List-II**.

List-I	List-II
A. Polyadelphous	I. Filaments of stamens united into more than two bundles
B. Syncarpous	II. Sterile stamen
C. Staminode	III. Stamens are attached to the petals
D. Epipetalous	IV. Carpels are fused

Choose the **correct** answer from the options given below :

- (1) A-II, B-I, C-III, D-IV (2) A-II, B-III, C-I, D-IV
- (3) A-I, B-II, C-IV, D-III (4) A-I, B-IV, C-II, D-III

Ans. (4)

Sol. NCERT-XI, Pg. # 64, 65

118. Identify the place(s) from the following where sacred groves are not found.

- A. Aravalli Hills of Rajasthan.
- B. Chanda and Bastar area of Madhya Pradesh.
- C. Mudumalai Sanctuary in Tamil Nadu.
- D. Khasi and Jaintia Hills in Meghalaya.

Choose the **correct** answer from the options given below :

- (1) C only (2) B and C only
- (3) C and D only (4) A only

Ans. (1)

Sol. NCERT-XII, Pg. # 225

119. 'Golden rice' is a genetically modified rice. It is an example of :

- (1) biofortification
- (2) pest-resistant crop
- (3) increased mineral usage efficiency
- (4) bioprospecting

Ans. (1)

Sol. NCERT-XII, Pg. # 179

120. The junction between ovule and funicle is represented by :

- (1) Integument (2) Chalaza
- (3) Hilum (4) Nucellus

Ans. (3)

Sol. NCERT-XII, Pg. # 9

121. Match **List-I** with **List-II**.

List-I	List-II
A. Toxin	I. Gum
B. Polymeric substance	II. Concanavalin A
C. Lectin	III. Ricin
D. Drug	IV. Vinblastin

Choose the **correct** answer from the option below :

- (1) A-II, B-IV, C-I, D-III (2) A-III, B-I, C-II, D-IV
- (3) A-II, B-I, C-IV, D-III (4) A-III, B-IV, C-II, D-I

Ans. (2)

Sol. NCERT-XI, Pg. # 108



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122. The asexual spores of Deuteromycetes are :

- (1) Conidia (2) Zoospores
(3) Basidiospores (4) Aplanospores

Ans. (1)

Sol. NCERT-XI, Pg. # 18

123. Cell wall formation in Bacteria is facilitated by :

- (1) Mesosomes (2) Golgi Apparatus
(3) Centrosomes (4) Ribosomes

Ans. (1)

Sol. NCERT-XI, Pg. # 91

124. Which of the following technique was used to elucidate the double helix model of DNA ?

- (1) Electromagnetic radiation
(2) UV-vis spectroscopy
(3) X-ray diffraction
(4) γ -radiation

Ans. (3)

Sol. NCERT-XII, Pg. # 81

125. Match **List-I** with **List-II**.

List-I	List-II
A. C_4 Pathway	I. $2H_2O \rightarrow 4H^+ + O_2 + 4e^-$
B. Light reaction	II. Ribulose 1,5-bisphosphate
C. Photorespiration	III. Phosphoenol Pyruvate
D. Calvin Cycle	IV. Phosphoglycolate

Choose the **correct** answer from the options given below :

- (1) A-IV, B-II, C-I, D-III (2) A-III, B-I, C-IV, D-II
(3) A-III, B-IV, C-II, D-I (4) A-II, B-I, C-IV, D-III

Ans. (2)

Sol. NCERT-XI, Pg. # 139-147

126. Which of the following represents the female gametophyte ?

- (1) Embryo sac (2) Nucellus
(3) Endosperm (4) Ovule

Ans. (1)

Sol. NCERT-XII, Pg. # 10

127. Which one of the following is not a complex polysaccharide ?

- (1) Chitin
(2) Glucosamine
(3) N-acetyl galactosamine
(4) Inulin

Ans. (4)

Sol. NCERT-XI, Pg. # 111

128. A yellow seeded and violet flowered pea plant was crossed with a green seeded and white flowered pea plant. The progeny showed plants with four different phenotypes. The yellow seeded and violet flowered parent must be:

- (1) Double heterozygous
(2) Homozygous for seed colour and heterozygous for flower colour
(3) Heterozygous for seed colour and homozygous for flower colour
(4) Double homozygous

Ans. (1)

Sol. NCERT-XII, Pg. # 64

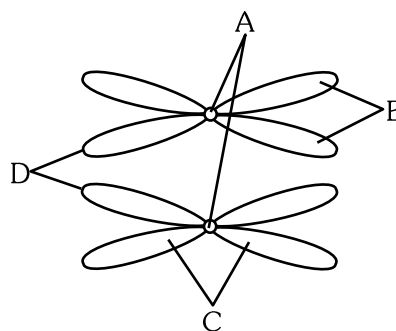
129. Which of the following is **not** the characteristic feature of genetic code?

- (1) The code is nearly universal
(2) The code has punctuations
(3) Some amino acids are coded by more than one codon, hence the code is degenerate
(4) The codon is triplet

Ans. (2)

Sol. NCERT-XII, Pg. # 96

130. Which of the following represent Sister chromatids in the given figure?



- (1) B (2) C
(3) D (4) A

Ans. (1)

Sol. NCERT-XI, Pg. # 127



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131. In plants, sucrose is converted into two molecule of monosaccharides due to action of invertase and then both enter the glycolytic pathway. Identify the monosaccharides.

- (1) Glucose and fructose
- (2) Glucose and galactose
- (3) Fructose and galactose
- (4) Glucose and glucose

Ans. (1)

Sol. NCERT-XI, Pg. # 156

132. A cross between tall plants was made resulting offsprings of tall and dwarf plants with ratio 3 : 1. The genotype (s) of both the parents are :

- (1) Tt and tt
- (2) Tt and Tt
- (3) tt and tt
- (4) TT and TT

Ans. (2)

Sol. NCERT-XII, Pg. # 57

133. How many molecules of ATP and NADPH are required respectively for fixation of every CO_2 molecule entering the Calvin cycle?

- (1) 4 and 2
- (2) 3 and 2
- (3) 2 and 3
- (4) 2 and 4

Ans. (2)

Sol. NCERT-XI, Pg. # 145

134. The photochemical phase of photosynthesis include:

- A. Oxygen release
- B. Formation of NADPH
- C. Use of ATP
- D. Water splitting
- E. Carboxylation

Choose the correct answer from the options given below :

- (1) A, B and D only
- (2) B, C and E only
- (3) A, B, C and D only
- (4) C and E only

Ans. (1)

Sol. NCERT-XI, Pg. # 138-140

135. Read the following statements and choose the set of **correct** statements :

- A. In cymose inflorescence, the flowers are borne in an acropetalous succession.
- B. In gulmohar, the flowers are actinomorphic.
- C. In the flowers of cucumber, the margin of the thalamus grows upward enclosing the ovary completely and getting fused with it, the other parts of flower arise above the ovary.
- D. Bracts are reduced leaves found at the base of the pedicel.
- E. The flowers are hypogynous in mustard.

Choose the **correct** answer from the options given below :

- (1) B, D, E only
- (2) A, C, D only
- (3) C, D, E only
- (4) A, B, C only

Ans. (3)

Sol. NCERT-XI, Pg. # 61, 62, 63

Botany : Section-B (Q. No. 136 to 150)

136. Given below are two statements :

Statement I : The rate of decomposition is not related to chemical composition of detritus and climatic factors.

Statement II : In a particular climatic conditions, decomposition rate is faster if detritus is rich in lignin and chitin.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are false.
- (2) **Statement I** is true but **Statement II** is false.
- (3) **Statement I** is false but **Statement II** is true.
- (4) Both **Statement I** and **Statement II** are true.

Ans. (1)

Sol. NCERT-XII, Pg. # 208



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137. Which of the following statements are **correct** about respiration?

- A.** Energy of oxidation - reduction is utilised for phosphorylation.
- B.** Oxygen acts as the final hydrogen acceptor.
- C.** The photo-oxidation energy is utilised for production of proton gradient required for phosphorylation.
- D.** The role of oxygen is limited to the terminal stage of the respiration process.
- E.** Protons cross the outer membrane of mitochondria through the channel formed by an integral membrane protein complex.

Choose the **correct** answer from the options given below:

- (1) A, B, D only (2) B, C, D, E only
- (3) A, C, D only (4) A, B, D, E only

Ans. (1)

Sol. NCERT-XI, Pg. # 160

138. Given below are two statements :

Statement I : Restriction endonuclease finds its specific recognition sequence and binds to the DNA.

Statement II : Restriction Endonuclease cuts each of the two strands of the double helix at specific points in their sugar phosphate backbone.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are false.
- (2) **Statement I** is true but **Statement II** is false.
- (3) **Statement I** is false but **Statement II** is true.
- (4) Both **Statement I** and **Statement II** are true.

Ans. (4)

Sol. NCERT-XII, Pg. # 166

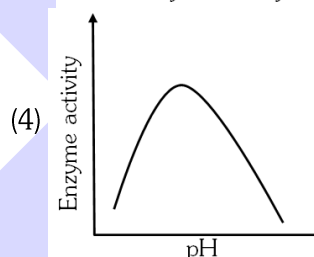
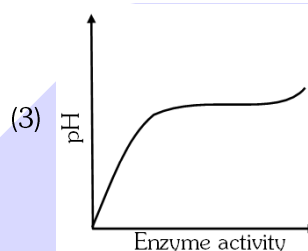
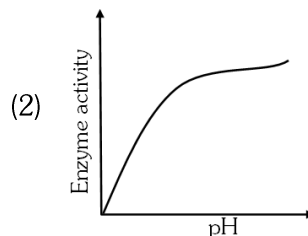
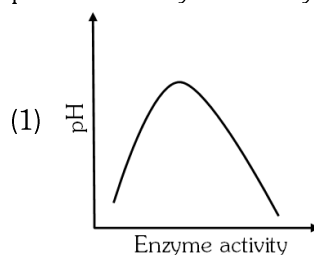
139. Which of the following is **not** an example of Mutualism?

- (1) Mycorrhizae
- (2) An orchid on a branch of mango plant
- (3) The wasp pollinating fig inflorescence
- (4) Lichens

Ans. (2)

Sol. NCERT-XII, Pg. # 201

140. Which of the following graphs explains the effect pH on the enzyme activity?



Ans. (4)

Sol. NCERT-XI, Pg. # 116

141. Read the following statements and find out the **correct** set of statements :

- A.** Companion cells help in maintaining the pressure gradient in the sieve tubes.
- B.** Gymnosperm lack vessels in their xylem.
- C.** The xylem vessels are devoid of cytoplasm.
- D.** Xylem fibres may be septate or aseptate.
- E.** A mature sieve element in phloem possesses cytoplasm, vacuole and nucleus.

Choose the **correct** answer from the options given below:

- (1) A, B, D, E only (2) A, B, C, D only
- (3) C, D, E only (4) B, C, D, E only

Ans. (2)

Sol. NCERT-XI, Pg. # 87 (Old NCERT)



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142. Highest annual Net Primary Productivity is observed in-

- (1) Temperate evergreen forest
- (2) Temperate deciduous forest
- (3) Tropical rain forest
- (4) Tropical deciduous forest

Ans. (3)

Sol. NCERT-XII, Pg. # 207

143. Given below are two statements :

Statement I : In collenchyma cell walls are thickened at corners due to deposition of cellulose, hemicellulose and pectin.

Statement II : Sclerenchyma consists of lignified cell walls and possesses pits.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are false.
- (2) **Statement I** is true but **Statement II** is false.
- (3) **Statement I** is false but **Statement II** is true.
- (4) Both **Statement I** and **Statement II** are true.

Ans. (4)

Sol. NCERT-XI, Pg. # 86 (Old NCERT)

144. Bt toxin Protein that can kill many insects, doesn't kill the *Bacillus* because:

- (1) *Bacillus* has protective mechanism against it.
- (2) In *Bacillus*, it is present in an inactive state and once insect ingests it, it is converted to active form.
- (3) The protoxin requires protein conjugate for its activation
- (4) *Bt* toxin requires acidic pH for activation

Ans. (2)

Sol. NCERT-XII, Pg. # 179

145. Which experimental material was used by Taylor and colleagues to prove that DNA in chromosomes replicates semiconservatively?

- (1) *Pisum sativum*
- (2) *Solanum tuberosum*
- (3) *Oryza sativa*
- (4) *Vicia faba*

Ans. (4)

Sol. NCERT-XII, Pg. # 90

146. Match List-I with List-II

List-I

- A. Collagen
- B. GLUT-4
- C. Trypsin
- D. RuBisCO

List-II

- I. Enzyme
- II. Most abundant enzyme in biosphere
- III. Most abundant protein in animal world
- IV. Enables glucose transport into cells.

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-III, B-IV, C-I, D-II
- (3) A-I, B-III, C-IV, D-II
- (4) A-III, B-I, C-IV, D-II

Ans. (2)

Sol. NCERT-XI, Pg. # 109

147. What is incorrect about ecosystem?

- (1) It may be anthropogenic in origin
- (2) It may be temporary or Permanent
- (3) It involves the function of flow of energy not recycling of nutrients.
- (4) It can vary from small sized pond to large sized sea.

Ans. (3)

Sol. NCERT-XII, Pg. # 205

148. The flowers of *Amorphophallus* provide safe place for the pollinator to lay eggs. This is an example of:

- (1) Anemophily
- (2) Floral reward
- (3) Commensalism
- (4) Amensalism

Ans. (2)

Sol. NCERT-XII, Pg. # 14



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149. Match List-I with List-II

List-I	List-II
A. Ethylene	I. Increasing length of stem
B. Cytokinins	II. Promotion of senescence and abscission
C. ABA	III. Delaying leaf senescence
D. Gibberellins	IV. Inhibition of seed germination

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
 (2) A-IV, B-II, C-I, D-III
 (3) A-III, B-IV, C-I, D-II
 (4) A-III, B-I, C-IV, D-II

Ans. (1)**Sol.** NCERT-XI, Pg. # 176-177150. In the *lac* operon the *i* gene code for :

- (1) repressor (2) β -galactosidase
 (3) permease (4) inducer

Ans. (1)**Sol.** NCERT-XII, Pg. # 101
Zoology : Section-A (Q. No. 151 to 185)
151. Name the hormone(s) **not** secreted by placenta.

- A. Estrogens
 B. Progestogens
 C. Relaxin and Inhibin
 D. hCG
 E. hPL

Choose the **correct** answer from the options given below:

- (1) C only (2) B only
 (3) A and C only (4) D and E only

Ans. (1)**Sol.** NCERT-XII, Pg. # 37

152. MTP (Medical Termination of Pregnancy) with oral pills is considered relatively safe upto-

- (1) 15-18 weeks of pregnancy
 (2) 20-24 weeks of pregnancy
 (3) 28-30 weeks of pregnancy
 (4) 7-9 weeks of pregnancy

Ans. (4)**Sol.** NCERT-XII, Pg. # 46

153. Match List-I with List-II:

List-I	List-II
A. Vertebrochondral Ribs	I. Ilium, Ischium, Pubis
B. Floating Ribs	II. 8 th , 9 th and 10 th Pairs
C. Scapula	III. Acromion
D. Coxal Bone	IV. 11 th and 12 th Pairs

Choose the **correct** answer from the options given below :

- (1) A-III, B-IV, C-II, D-I (2) A-II, B-IV, C-III, D-I
 (3) A-I, B-IV, C-II, D-III (4) A-I, B-III, C-II, D-IV

Ans. (2)**Sol.** NCERT-XI, Pg. # 225, 226

154. Match List-I with List-II:

List-I	List-II
A. Parathyroid hormone	I. Flight or fight response
B. Epinephrine and Nor-epinephrine	II. Regulates the body's biological clock
C. Thyroid stimulating hormone	III. Increases blood Ca^{2+} level
D. Melatonin	IV. Synthesis of T_3 and T_4 hormone

Choose the **correct** answer from the options given below :

- (1) A-III, B-IV, C-I, D-II (2) A-III, B-I, C-II, D-IV
 (3) A-I, B-II, C-III, D-IV (4) A-III, B-I, C-IV, D-II

Ans. (4)**Sol.** NCERT-XI, Pg. # 241-244

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155. Match List-I with List-II:

List-I	List-II
A. <i>Saccharomyces cerevisiae</i>	I. Citric acid
B. <i>Aspergillus niger</i>	II. Butyric acid
C. <i>Trichoderma polysporum</i>	III. Ethanol
D. <i>Clostridium butylicum</i>	IV. Cyclosporin-A

Choose the correct answer from the options given below :

- (1) A-IV, B-II, C-III, D-I
- (2) A-III, B-IV, C-I, D-II
- (3) A-I, B-III, C-II, D-IV
- (4) A-III, B-I, C-IV, D-II

Ans. (4)

Sol. NCERT-XII, Pg. # 153

156. Given below are two statements:

Statement-I : The ground substances of bone is solid, pliable and resist compression.

Statement-II : The bone marrow of all bones is site of production of blood cells.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both **Statement-I** and **Statement-II** are incorrect.
- (2) **Statement-I** is correct but **Statement-II** is incorrect.
- (3) **Statement-I** is incorrect but **Statement-II** is correct.
- (4) Both **Statement-I** and **Statement-II** are correct.

Ans. (1)

Sol. NCERT-XI, Pg. # 104 (Old NCERT)

157. Which of the following is NOT an example of assisted reproductive technology (ART) ?

- (1) GIFT
- (2) IUI
- (3) IUD
- (4) IVF

Ans. (3)

Sol. NCERT-XII, Pg. # 48

158. Given below are two statements:

Statement-I : Aphids and mosquitoes are natural pests.

Statement-II : Lady bird and dragonflies are natural pests controllers.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both **Statement-I** and **Statement-II** are incorrect.
- (2) **Statement-I** is correct but **Statement-II** is incorrect.
- (3) **Statement-I** is incorrect but **Statement-II** is correct.
- (4) Both **Statement-I** and **Statement-II** are correct.

Ans. (4)

Sol. NCERT-XII, Pg. # 157

159. Given below are two statements:

Statement-I : Bacterial infection by *Bacillus thuringiensis* can kill only caterpillar larvae.

Statement-II : *B. thuringiensis* cannot kill adult moths.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both **Statement-I** and **Statement-II** are incorrect.
- (2) **Statement-I** is correct but **Statement-II** is incorrect.
- (3) **Statement-I** is incorrect but **Statement-II** is correct.
- (4) Both **Statement-I** and **Statement-II** are correct.

Ans. (4)

Sol. NCERT-XII, Pg. # 157



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160. Given below is a list of some sexually reproducing animals. Select the hermaphrodite animals from the list.

- A. Cockroach B. Leech
C. Housefly D. Tapeworm
E. Earthworm

Choose the **correct** answer from the option given below:

- (1) C and D only (2) B, D and E only
(3) A, C and E only (4) A, B and D only

Ans. (2)

Sol. NCERT-XI, Pg. # 42, 43, 44

161. Arrange in a proper sequence the pathway of sperms from testis to outside in human male reproductive system.

- A. Vas deferens B. Rete testis
C. Seminiferous tubules D. Vasa efferentia
E. Urethra F. Epididymis

Choose the **correct** answer from the option given below:

- (1) C → A → B → D → F → E
(2) B → A → D → C → F → E
(3) C → B → D → F → A → E
(4) F → A → B → D → E → C

Ans. (3)

Sol. NCERT-XII, Pg. # 32

162. The catalytic cycle of an enzyme action is described as:

- A. Enzyme releases products of the reaction and gets free.
B. Substrate induces the enzymes to alter its shape.
C. The substrate binds with the active site of the enzymes.
D. Enzymes-product complex is formed.

Choose the **correct** answer from the options given below:

- (1) C → B → D → A (2) B → C → D → A
(3) D → C → A → B (4) C → B → A → D

Ans. (1)

Sol. NCERT-XI, Pg. # 115

163. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Connell's elegant field experiment showed that on the rocky sea, the larger and competitively superior barnacles dominate the intertidal area, and excludes the smaller barnacle from that area.

Reason R : Generally herbivores and plants appear to be more adversely affected by competition than Carnivores.

In the light of the above statements, choose the **correct** answer from the option given below :

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
(2) **A** is true but **R** is false.
(3) **A** is false but **R** is true.
(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Ans. (1)

Sol. NCERT-XII, Pg. # 199

164. Match **List-I** with **List-II**:

List-I	List-II
A. Ampicillin	I. Restriction Sites
B. Ori	II. Blue/White screening
C. Multiple Cloning Sites	III. Selectable marker
D. β galactosidase gene	IV. Sequence from where replication starts

Choose the **correct** answer from the options given below :

- (1) A-III, B-I, C-IV, D-II (2) A-IV, B-III, C-II, D-I
(3) A-II, B-IV, C-I, D-III (4) A-III, B-IV, C-I, D-II

Ans. (4)

Sol. NCERT-XII, Pg. # 169, 170



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165. Match List-I with List-II:

List-I	List-II
A. Emphysema	I. Proliferation of fibrous tissues
B. Asbestosis	II. Alveolar walls are damaged and respiratory surface is decreased
C. Asthma	III. Serious lung damage
D. Fibrosis	IV. Difficulty in breathing due to inflammation of bronchi and bronchioles

Choose the **correct** answer from the options given below :

- (1) A-II, B-III, C-IV, D-I (2) A-IV, B-III, C-II, D-I
(3) A-I, B-II, C-IV, D-III (4) A-I, B-III, C-IV, D-II

Ans. (1)

Sol. NCERT-XI, Pg. # 190, 191

166. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : In honey bee population, sons do not have fathers but have grandfathers.

Reason R : Haploidy in drones of honey bee is due to parthenogenesis while fertilisation results into female bees.

In the light of the above statements, choose the **correct** answer from the option given below :

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
(2) **A** is true but **R** is false.
(3) **A** is false but **R** is true.
(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Ans. (4)

Sol. NCERT-XII, Pg. # 71

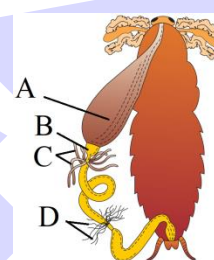
167. Which of the following is 'NOT' an autosomal recessive disorders ?

- (1) Phenylketonuria
(2) Haemophilia
(3) Thalassemia
(4) Sickle cell anemia

Ans. (2)

Sol. NCERT-XII, Pg. # 74, 75

168. In the following diagram of alimentary canal of cockroach identify the labels A, B, C and D respectively.



- (1) A-Oesophagus, B-Crop, C-Salivary glands, D-Hepatic caeca
(2) A-Crop, B-Gizzard, C-Hepatic caeca, D-Malpighian tubules
(3) A-Gizzard, B-Hepatic caeca, C-Malpighian tubules, D-Cilia
(4) A-Gizzard, B-Crop, C-Malpighian tubules, D-Hepatic caeca

Ans. (2)

Sol. NCERT-XI, Pg. # 113 (Old NCERT)

169. Which of the following group of animals have three chambered heart ?

- (1) Chameleon, Hyla, Calotes
(2) Crocodilus, Pavo, Clarias
(3) Exocoetus, Hemidactylus, Rana
(4) Bufo, Alligator, Struthio

Ans. (1)

Sol. NCERT-XI, Pg. # 48, 49



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170. Match **List-I** with **List-II**:

List-I	List-II
A. Ribozyme	I. Glucose transport
B. Lecithin	II. Non proteinaceous Enzyme
C. GLUT-4	III. Lipid
D. Vitamins	IV. Coenzyme

Choose the **correct** answer from the options given below :

- (1) A-IV, B-III, C-II, D-I (2) A-II, B-I, C-III, D-IV
(3) A-II, B-III, C-I, D-IV (4) A-IV, B-III, C-I, D-II

Ans. (3)**Sol.** NCERT-XI, Pg. # 106, 109, 113, 118

171. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Some ethical standards are required to evaluate the morality of scientific human activities that might help or harm living organisms.

Reason R : Genetic Engineering Approval Committee make decisions regarding the validity of Genetically Modified (GM) research and the safety of introducing GM organisms for public services.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both **A** and **R** are correct but **R** is NOT the correct explanation of **A**.
(2) **A** is correct but **R** is not correct.
(3) **A** is not correct but **R** is correct.
(4) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

Ans. (1)**Sol.** NCERT-XII, Pg. # 185

172. Use of hyper immune horse serum raised against a specific pathogen, given to a recipient is an example of:

- (1) Artificial active immunity
(2) Natural passive immunity
(3) Artificial passive immunity
(4) Natural active immunity

Ans. (3)**Sol.** NCERT-XII, Pg. # 136173. Match **List-I** with **List-II**:

List-I Cell Cycle	List-II Events
A. G ₁ Phase	I. Cell is metabolically active but no proliferation until required
B. S Phase	II. Replication of DNA
C. G ₀ Stage	III. Synthesis of Proteins
D. G ₂ Phase	IV. Cell is metabolically active and grows continuously

Choose the **correct** answer from the options given below :

- (1) A-IV, B-II, C-I, D-III (2) A-II, B-IV, C-I, D-III
(3) A-I, B-III, C-II, D-IV (4) A-I, B-II, C-IV, D-III

Ans. (1)**Sol.** NCERT-XI, Pg. # 121, 122174. Match **List-I** with **List-II**:

List-I	List-II
A. DNA	I. M. Meselson and F. Stahl
B. <i>Pneumococcus</i>	II. A. Harshey and M. Chase
C. <i>E. coli</i>	III. F. Griffith
D. Bacteriophage	IV. Alec Jeffreys

Choose the **correct** answer from the options given below :

- (1) A-IV, B-III, C-I, D-II (2) A-II, B-III, C-I, D-IV
(3) A-III, B-II, C-I, D-IV (4) A-IV, B-III, C-II, D-I

Ans. (1)**Sol.** NCERT-XII, Pg. # 84, 85, 89, 105175. Match **List-I** with **List-II**:

List-I (Biological Name)	List-II (Common Name)
A. <i>Asterias</i>	I. Sea urchin
B. <i>Antedon</i>	II. Brittle star
C. <i>Echinus</i>	III. Sea lily
D. <i>Ophiura</i>	IV. Star fish

Choose the **correct** answer from the options given below :

- (1) A-IV, B-III, C-I, D-II (2) A-II, B-III, C-I, D-IV
(3) A-I, B-II, C-IV, D-III (4) A-III, B-IV, C-II, D-I

Ans. (1)**Sol.** NCERT-XI, Pg. # 45

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176. The animal coelacanth caught in South Africa is an example of vertebrate transition from:

- (1) Amphibia to Reptilia
- (2) Reptilia to Bird
- (3) Reptilia to Mammal
- (4) Fish to Amphibia

Ans. (4)

Sol. NCERT-XII, Pg. # 122

177. Match **List-I** with **List-II**:

List-I	List-II
A. Tricuspid valves	I. Guards the opening between right atrium and right ventricle
B. Sino-atrial node	II. Guards the opening between the left atrium and the left ventricle
C. Mitral valves	III. Guards the opening of right and left ventricles into pulmonary artery and aorta, respectively
D. Semilunar valves	IV. Cardiac musculature in the right upper corner of right atrium

Choose the **correct** answer from the options given below :

- (1) A-II, B-I, C-IV, D-III (2) A-III, B-IV, C-I, D-II
(3) A-I, B-IV, C-II, D-III (4) A-III, B-IV, C-II, D-I

Ans. (3)

Sol. NCERT-XI, Pg. # 198, 199

178. Identify the correct statements about Genetic codons.

- A. AUG is initiator codon and codes for Glycine
- B. UAA codes for Tyrosine
- C. The codons are mostly universal
- D. UAG is terminator codon
- E. More than one codons code for single amino acid

- (1) A, D, E (2) C, D, E (3) A, B, C (4) C, B, D

Ans. (2)

Sol. NCERT-XII, Pg. # 96

179. Which part of human nephron is impermeable to water?

- (1) Distal convoluted tubule
- (2) Ascending limb of loop of Henle
- (3) Descending limb of loop of Henle
- (4) Proximal convoluted tubule

Ans. (2)

Sol. NCERT-XI, Pg. # 209

180. In standard ECG diagram, the P-wave represents:

- (1) Repolarisation of the ventricle
- (2) Depolarisation of the ventricle
- (3) End of systole
- (4) Depolarisation of the atria

Ans. (4)

Sol. NCERT-XI, Pg. # 201

181. A Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: The interaction, in which one species benefits and the other is neither harmed nor benefited, is known as commensalism.

Reason R: Egrets always forage close to where the cattle are grazing, otherwise it is difficult for the egrets to find the insect and catch.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both A and R are true but R is NOT the correct explanation of A.
- (2) A is true but R is false.
- (3) A is false but R is true.
- (4) Both A and R are true and R is the correct explanation of A.

Ans. (1)

Sol. NCERT-XII, Pg. # 201



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182. Match List-I with List-II relating to major features of embryonic development at different timings of pregnancy:

List-I (No. of Weeks)	List-II (Major features)
A. After 4 weeks of pregnancy	I. Eye-lids separate and eyelashes are formed
B. After 12 weeks of pregnancy	II. Heart is formed
C. After 24 weeks of pregnancy	III. Limbs and digits are developed
D. After 8 weeks of pregnancy	IV. External genital organs are well developed

Choose the correct answer from the options given below :

- (1) A-II, B-I, C-III, D-IV
 (2) A-II, B-IV, C-I, D-III
 (3) A-III, B-IV, C-II, D-I
 (4) A-III, B-IV, C-I, D-II

Ans. (2)

Sol. NCERT-XII, Pg. # 38

183. Match List-I with List-II

List-I	List-II
A. Limbic system	I. Body temperature
B. Arachnoid	II. Tract of nerve fibres
C. Corpus callosum	III. Amygdala
D. Hypothalamus	IV. Cranial meninges

Choose the correct answer from the options given below :

- (1) A-I, B-II, C-IV, D-III
 (2) A-II, B-I, C-III, D-IV
 (3) A-IV, B-II, C-III, D-I
 (4) A-III, B-IV, C-II, D-I

Ans. (4)

Sol. NCERT-XI, Pg. # 235, 236

184. Match List-I with List-II

List-I	List-II
A. Tertiary consumer	I. Grass
B. Secondary consumer	II. Lion
C. Primary consumer	III. Wolf
D. Primary producer	IV. Goat

Choose the correct answer from the options given below :

- (1) A-III, B-IV, C-II, D-I
 (2) A-III, B-II, C-IV, D-I
 (3) A-II, B-III, C-IV, D-I
 (4) A-I, B-II, C-III, D-IV

Ans. (3)

Sol. NCERT-XII, Pg. # 210

185. Match List-I with List-II

List-I	List-II
A. Cilia	I. Spindle fibres
B. Endoplasmic Reticulum	II. Cristae
C. Mitochondria	III. Axoneme
D. Kinetochore	IV. Ribosomes

Choose the correct answer from the options given below :

- (1) A-III, B-IV, C-II, D-I
 (2) A-I, B-III, C-II, D-IV
 (3) A-III, B-II, C-I, D-IV
 (4) A-I, B-IV, C-II, D-III

Ans. (1)

Sol. NCERT-XI, Pg. # 95, 97, 99, 101

Zoology : Section-B (Q. No. 186 to 200)

186. The manifestations of phenylketonuria are given below. Choose the **incorrect** answer from the given options :

- (1) Skin rashes
 (2) Excessive accumulation of phenylpyruvate in blood
 (3) Furrowed tongue
 (4) Mental retardation

Ans. (3)

Sol. NCERT-XII, Pg. # 75



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187. In human the process of respiration involves the following steps.

- A. Diffusion of gases across alveolar membrane
- B. Diffusion of gases between blood and tissues
- C. Transport of gases by blood
- D. Utilisation of O_2 by the cells for catabolic reactions
- E. Breathing or pulmonary ventilation

Choose the correct sequence of steps from the options given below:

- (1) $C \rightarrow E \rightarrow A \rightarrow B \rightarrow D$
- (2) $B \rightarrow D \rightarrow C \rightarrow E \rightarrow A$
- (3) $E \rightarrow A \rightarrow C \rightarrow B \rightarrow D$
- (4) $A \rightarrow E \rightarrow B \rightarrow C \rightarrow D$

Ans. (3)

Sol. NCERT-XI, Pg. # 185

188. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: After death, the body muscles get stiff as a state of partial contraction of muscles.

Reason R: Attachment of myosin head to unexposed sites on actin can take place partially.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both **A** and **R** are correct but **R** is NOT the correct explanation of **A**.
- (2) **A** is correct but **R** is not correct.
- (3) **A** is not correct but **R** is correct.
- (4) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

Ans. (2)

Sol. Module

189. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Some human organs like liver, kidney fail to function normally and transplantation is the only remedy to enable the patient to live a normal life.

Reason R : Tissue matching and blood group matching are essential before undertaking any transplant and patient has to take immuno-suppressant thereafter.

In the light of the above statements, **choose** the **correct** answer from the options given below:

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (2) **A** is true but **R** is false.
- (3) **A** is false but **R** is true.
- (4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Ans. (1)

Sol. NCERT-XII, Pg. # 135, 136

190. With regard to Hormones, identify the **correct** statements.

- A. Epinephrine is a peptide hormone.
- B. Progesterone is a peptide hormone.
- C. Hormones that interact with membrane bound receptors normally do not enter target cell, but generate second messengers.
- D. Hormones that interact with intracellular receptors mostly regulate gene expression.
- E. Insulin is a amino acid derivative hormone.

Choose the **most appropriate** answer from the options given below:

- (1) C and E only
- (2) C and D only
- (3) B and C only
- (4) A and B only

Ans. (2)

Sol. NCERT-XI, Pg. # 248



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191. In animal kingdom along with XX-XY mechanism, many other variants of chromosomal sex determination mechanism have been described such as XX-XO mechanism, ZZ-ZW mechanism ZZ-ZO mechanism and haplo-diploidy mechanism, which are present, respectively in:

- (1) Grasshopper, Birds, Moth, Wasp
- (2) Birds, Moth, Grasshopper, Wasp
- (3) Grasshopper, Moth, Birds, Wasp
- (4) Birds, Grasshopper, Wasp, Moth

Ans. (1)

Sol. NCERT-XII, Pg. # 69, 70

192. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : In hemichordates the body cavity is lined by mesoderm.

Reason R : In aschelminthes the body cavity is not lined by mesoderm.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (2) **A** is true but **R** is false.
- (3) **A** is false but **R** is true.
- (4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Ans. (1)

Sol. NCERT-XI, Pg. # 43, 45

193. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**. **Assertion A :** Juxta Glomerular Apparatus (JGA) plays an important role in regulation of glomerular filtration rate (GFR).

Reason R : A fall in GFR can activate the J.G. cells to release renin which can stimulate the glomerular blood flow and there by bringing the GFR to normal level.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (2) **A** is true but **R** is false.
- (3) **A** is false but **R** is true.
- (4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Ans. (4)

Sol. NCERT-XI, Pg. # 212

194. Assuming Hardy Weinberg's Principle, in certain population the frequencies of three genotypes are as follows:

Genotypes	AA	Aa	aa
Frequency	22%	62%	16%

What is likely frequency of 'A' and 'a' alleles ?

- (1) 40% and 60%
- (2) 50% and 50%
- (3) 30% and 70%
- (4) 53% and 47%

Ans. (4)

Sol. NCERT-XII, Pg. # 121

195. Given below are two statements:

Statement I : The Golgi cisternae are concentrically arranged near the nucleus with distinct convex *cis* or the forming face and concave *trans* or the maturing face.

Statement II : The *cis* and *trans* faces of the organelle are identical and interconnected.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are false.
- (2) **Statement I** is true but **Statement II** is false.
- (3) **Statement I** is false but **Statement II** is true.
- (4) Both **Statement I** and **Statement II** are true.

Ans. (2)

Sol. NCERT-XI, Pg. # 95, 96

196. Given below are two statements:

Statement I : Degeneration of corpus luteum in the absence of fertilization is the cause for disintegration of endometrium.

Statement II : Cyclic menstruation indicates normal reproductive phase in human females and extends between menarche and menopause.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are false
- (2) **Statement I** is true but **Statement II** is false.
- (3) **Statement I** is false but **Statement II** is true.
- (4) Both **Statement I** and **Statement II** are true.

Ans. (4)

Sol. NCERT-XII, Pg. # 35



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197. Match **List-I** with **List-II**

List-I (Events)	List-II (Prophase-I)
A. Chromosomes start pairing together to form synaptonemal complex	I. Pachytene
B. Chromosomes visible under light microscope and compaction continues	II. Zygotene
C. Four chromatids of bivalent chromosomes become distinct and recombinant nodules appear	III. Diplotene
D. Dissolution of synaptonemal complex	IV. Leptotene

Choose the **correct** answer from the options given below :

- (1) A-I, B-IV, C-III, D-II
- (2) A-III, B-II, C-IV, D-I
- (3) A-II, B-IV, C-I, D-III
- (4) A-II, B-I, C-IV, D-III

Ans. (3)

Sol. NCERT-XI, Pg. # 126

198. Match **List-I** with **List-II**

List-I (Cockroach)	List-II (Term used)
A. Metamorphosis	I. Ommatidia
B. Brain	II. Paurometabolous
C. Vision	III. Spiracles
D. Respiration	IV. Supra-oesophageal ganglion

Choose the **correct** answer from the options given below :

- (1) A-I, B-III, C-IV, D-II
- (2) A-III, B-II, C-I, D-IV
- (3) A-II, B-IV, C-I, D-III
- (4) A-II, B-I, C-IV, D-III

Ans. (3)

Sol. NCERT-XI, Pg. # 113, 114, 115 (Old NCERT)

199. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : There is always a sharp decline in dissolved oxygen downstream from the point of sewage discharge, which causes mortality of fish and other aquatic creatures.

Reason R : Micro-organisms involved in bio-degradation of organic matter in the receiving water body consumes a lot of oxygen.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (2) **A** is true but **R** is false.
- (3) **A** is false but **R** is true.
- (4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Ans. (4)

Sol. NCERT-XII, Pg. # 274, 275 (Old NCERT)

200. The epithelial tissue has the following features:

- A. Bear cilia on their free surface.
- B. Provides a lining for some parts of body.
- C. Compound epithelium is multilayered meant for limited role in secretion and absorption.
- D. Located in the tip of nose.
- E. Exocrine glands possess epithelial tissue.

Choose the **most appropriate** answer from the options given below:

- (1) A, B and C only
- (2) B, C and E only
- (3) C, D and E only
- (4) B, D and E only

Ans. (2)

Sol. NCERT-XI, Pg. # 101, 102 (Old NCERT)



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